PHILIPPINE ARMY MANUAL 3-03

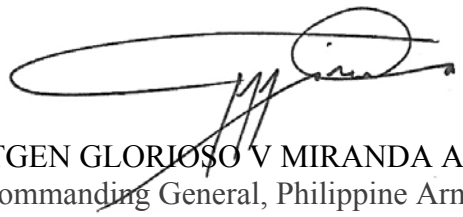
COMBINED ARMS OPERATIONS

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LTGEN GLORIOSO V MIRANDA AFP
Commanding General, Philippine Army

Preface

1. In order to holistically achieve the Philippine Army's vision of becoming "a world-class Army that is a source of national pride", it is imperative that the Army continuously develop the capability of its forces to be able to keep pace and be adaptable to the fast changing operational environment and to effectively execute its mandate as a dependable ground force.

2. One of the emphases given by Commanding General, Philippine Army is to develop the Philippine Army's capability in combined arms operations. He pointed out that the successful conduct of operations hinge on the proper force mix and employment of Army capabilities.

3. Likewise, as outlined in the Philippine Army Operating Concept and the Landpower Maneuver Concept, central to the Philippine Army fighting capability is the conduct of combined arms operations. It is a distinctive competency of the Philippine army that applies its different combat and combat support capabilities in the conduct of military affairs across the range of all Army operations. But because of the absence of doctrine, the critical capability of PA forces to plan for and conduct combined arms operations have not been fully entrenched in the Army. It is therefore imperative for the Army to develop this Combined Arms Operations doctrine as this will serve as guide for the commanders in the planning, integrating, synchronizing and employment of a proper force mix capable of maximizing all warfighting functions and rapidly accomplish he mission in any given environment. The manual will also provide useable guides for training planners and managers on the conduct of combined arms training exercise that will develop PA units into a combined arms capable fighting force.

4. Stakeholders of this Combined Arms Operations manual are the Army field commanders, combat support and combat service support units, PA Education and Training institutions, other AFP units directly involved in operations and individuals involved in the planning and execution of joint and combined arms operations.

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Chapter I
Introduction to Combined Arms Operations

0101. **Combined Arms Operations** is an approach to warfare which seeks to integrate the different capabilities of an army to achieve mutually complementary effects capable of overwhelming enemy forces. Combined arms hit the enemy with two or more capabilities simultaneously in such a manner that defense from one capability makes the enemy vulnerable to the other. The shock and awe produced by the simultaneous use of multiple capabilities overwhelm the enemy through lethal and non-lethal means thereby reducing the enemy's will and capacity to fight, ultimately leading to mission accomplishment of friendly troops. It is the synchronization, integration, and seamless execution of capabilities to attain desired effects.

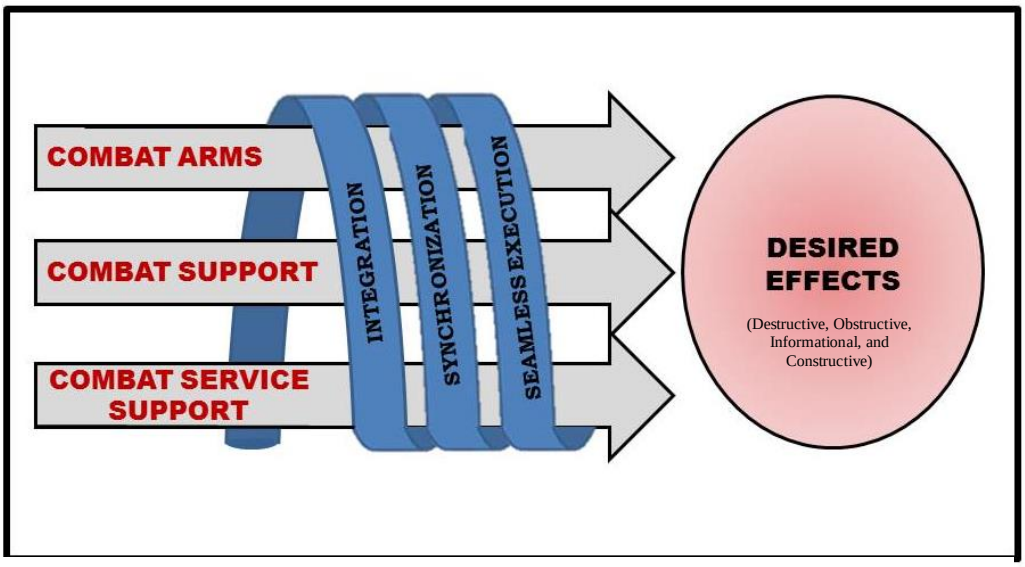


Figure 1.1 - Combined Arms Operations Illustrated

0102. Lower-echelon units of a combined arms team may be of similar types, a balanced mixture of such units are combined into an effective higher-echelon unit, whether formally in a table of organization or informally in an *ad hoc* solution to a battlefield problem. For example, a mechanized infantry brigade is a classic example of a CATms that consists of a mixture of mechanized infantry, infantry, artillery, reconnaissance units, and perhaps even combat aviation units, all coordinated and directed by a unified command structure.

0103. Combined Arms Operations is not a new concept in warfare. For centuries, armies have been using different kinds of combat arms to defeat their enemies and triumph in battle. Skirmishers, cavalry, foot soldiers, and artillery have been used in combination with one another to impose a nation's will on another or defend their territory. World War I saw the advent of a new form of cavalry through tanks that crossed the trenches in Europe and brought an end to the stalemate of trench warfare. World War II would make more evident the use of Combined Arms Operations particularly Germany's blitzkrieg warfare which combined armor, infantry, and motorized infantry supported by artillery and aircraft that brought much of mainland Europe to its knees.

0104. The wars and conflicts that followed the Second World War would see to it that armies continued the development of their combined arms doctrines and tactics.

Developments in military technology has produced more types of capabilities that can be used in Combined Arms Operations, however the basic principle remains to be the same, that is, Combined Arms Operations utilizes different capabilities to be triumphant over an enemy.

Combined Arms Operations in Philippine Settings

0105. Combined Arms Operations is not new for the Philippine Army. Throughout its history, the Philippine Army executed operations by employing two or more capabilities in numerous instances. However, because of the absence of doctrine, Combined Arms Operations is not fully entrenched in the Army's system. Numerous major events highlighted the employment of Combined Arms Operations in the Philippine Army.

0106. In War Plan Orange 3 during World War II, combined arms capabilities were utilized during the defense of Bataan and Corregidor. American and Filipino units such as infantry, cavalry, tank, field artillery, and engineer regiments, along with the quartermaster and medical corps among others, collectively contributed in the defense of the Philippines and delayed the timeline of Japanese forces for almost four (4) months. The Japanese invaders also utilized combined arms against the Filipino and American defenders. The assault on Bataan was a combination of infantry, armor, artillery, and supported by aircraft. With the fall of Bataan, Corregidor was bombarded by a combination of artillery and aircraft bombing and strafing runs. The inclusion of armor assets as part of the assault force at Corregidor changed the tide of battle. This was complicated by the lack of anti-armor capability of the defenders of Corregidor. In the end, the combination of Japanese infantry and armor forced the surrender of Corregidor and the cessation of organized defense for the Philippines.

0107. The most classic example of Combined Arms Operations by the Philippine Army was exemplified by the Philippine Expeditionary Force to Korea (PEFTOK) when a total of five (5) Battalion Combat Teams (BCT) were sent during the Korean War from 1950 to 1955. The BCTs are highly mobile, compact, and self-supporting battalion-sized fighting units designed to operate independently of each other in their area of operations.

0108. Each BCT, called the Philippine or Filipino Battalion, consisted of three rifle companies, a heavy weapons company, a headquarters company, and a 105mm howitzer company. The 10th BCT (Motorized), the first BCT to be deployed and only PEFTOK BCT with an armor component, had a company of 29 M4 Sherman Tanks and a company of M5 Stuart Light Tanks.

0109. In its campaign against secessionist and terrorist groups in Southern Philippines, the Philippine Army often employs multiple capabilities to accomplish its mission. A combination of infantry, mechanized infantry, and light armor units conduct operations as part of a larger force. Artillery fires, close air support and sometimes naval gunfire provide fire support for maneuver forces. Likewise, intelligence, surveillance, and reconnaissance provide essential information on the picture of the battlefield. Sustainment units provide logistics, personnel, and medical services to fighting units while CMO units projects the legitimacy of the Army's actions and provides liaising to local authorities, local and national line agencies, non-governmental organizations, and international aid agencies.

0110. In the Army's campaign against secessionist and terrorist groups several operations proved the importance of combined arms namely in the Reina Regente Campaign; Battle of Malitubog-Maridagao Complex; Battle of Dunguan, Aleosan, North Cotabato; Battle of

Buldon, Maguindanao; Battle of Rajah Muda; the fall of Camp Abubakr, Maguindanao; and, Butig Campaign, Lanao Del Sur.

0111. In the campaign against the CPP-NPA-NDF (CNN), the Philippine Army utilized its different capabilities in its objective of eliminating CNN influence and combatants in the provinces. The CNN as, a different threat group from the secessionist and terrorist groups called for a different strategy that gave focus on winning the hearts and minds of the populace. This difference gave attention to efforts that support national development in order to address root causes of the insurgency. Combat operations against the CNN saw the utilization of SOF units supported by infantry, armor, and artillery assets, complemented by intelligence units, which resulted in the elimination of CNN strongholds. A case in point is the operations against the CNN stronghold in Marag Valley, Apayao during the height of insurgency in the late 80's and the recent capture of the main camp of the Eastern Visayas Regional Part Committee in 2010 which saw the employment of SOF back by infantry and armor units and supported by artillery and air support in a coordinated and synchronized manner.

0112. The Zamboanga City Crisis in September 2013 presents one of the most recent Combined Arms Operations done by the Philippine Army. During the 21-day battle, the Army employed a combination of infantry and armor units to engage the rouge MNLF fighters in Zamboanga City. Special Operations Forces were however heavily utilized during the battle. Fire support was limited to indirect fire support provided by mortar teams and armor units had limited firepower in terms of penetrating the concrete cover and target acquisition capabilities to effectively engage enemy units in the urban environment. Armor assets were utilized in establishing foothold, providing armor protection, and providing suppressive cover during the insertion of SOF and infantry units in areas occupied by the enemy. Intelligence units provided real time intelligence utilizing unmanned aerial system. CMO units conducted loud speaker operations and psychological operations to dissuade the enemy, rally popular support, and establish the legitimacy of the military action. Sustainment units provided logistical and medical support to troops. Overall command and control was provided by the 1st Infantry Division.

0113. Philippine Army has been performing a non-traditional role of conducting Humanitarian Assistance and Disaster Response operations due its nature of having the necessary organization, training, personnel, and equipment that can respond to disasters. This stems from the fact that military assets can not only be employed for combat purposes but can also be utilized in disaster response and rehabilitation operations. For instance, in the aftermath of Typhoon Yolanda there was a need for the integration and synchronization of the different capabilities of the Army and AFP in general in providing support to civil authorities. The need for combined arms was highlighted in the exigency for the Army to provide area security to restore law and order in the affected areas and support movement of relief goods to include the security of such supplies. Also highlighted was the need to establish lines of communication particularly during the first few hours after the landfall of Typhoon Yolanda in Tacloban. Army engineers also played an important role in clearing road nets and establishing alternate supply routes to immediately access and provide relief goods to isolated areas. Likewise, K9 elements painstakingly conducted search and rescue operations and were instrumental to the recovery of numerous victims. Lastly, the establishment of a better picture of the operating environment was needed to assess which areas are to be prioritized in terms of providing relief. While non-combat in nature, the

utilization and synchronization of the different capabilities proved to be vital in the success of the operations after Typhoon Yolanda.

The Operational Environment and Its Effects to Combined Arms Operations

0114. The operational environment is a composite of the conditions, circumstances, and other factors that affect the employment of capabilities and bear on the decisions of the commander. Army leaders plan, prepare, execute, and assess operations by analyzing the operational environment in terms of the operational variables and mission variables. The operational variables consist of political, military, economic, social, information, infrastructure, physical environment, and time (PMESII-PT). On the other hand, mission variables consist of mission, enemy, terrain and weather, troops and support available, time available, civil considerations (METT-TC). The interaction of these variables in a specific situation, domain (land, maritime, air, space, or cyberspace), area of operations, or area of interest describes a commander's operational environment but does not limit it. Operational environments are not static. Within an operational environment, an Army leader may conduct major combat, military engagement and humanitarian assistance simultaneously. Army doctrine has always stated that Army forces must be prepared to transition rapidly from one type of operation to another. Army forces simultaneously and continuously combine offensive, defensive, and support to civil authority through a blend of Combined Arms Operations and security operations.

0115. The Philippine Army's operational environment has the following physical and non-physical characteristics that impacts the employment of combined arms capabilities:

a. **Archipelagic** – The Philippines consists of around 7, 100 islands. Every island has a unique physical characteristics ranging from vast flat terrain to mountainous terrain and long coastlines. The variable size and resources of each island allows for long-term habitation and development. The characteristics of these physical operating environments dictate that most military operations are affected and influenced by the archipelagic nature of the Philippines with four (4) distinct physical domains: land domain, maritime domain, air domain, and cyberspace domain. Emphasis is given on the land domain, the area of the Earth's surface ending at the water mark and overlapping with the maritime domain in the landward segment of the littorals, as this where Combined Arms Operations is most likely to take place. The employment of CATms in such archipelagic nature requires vast amount of logistics. Considerations must be included in terms of rapidly transporting units that can be as large as Brigade Combat Team either through the air or sea. For example, terrain features may restrict inland transportation and mobility corridors. While in the case of water transportation, crossing rivers are challenges that can be overcome through appropriate bridging equipment and techniques. Inter-island crossing on the other hand may require the lift capability of the Air Force and the Navy.

b. **Disaster-Prone** – Its geographic location along the "Pacific Ring of Fire" and "Pacific Typhoon Belt" makes the Philippines prone to natural calamities. With this, the Philippine Army's role in HADR will continue to compete with its traditional role of warfighting. The disaster prone nature of the Philippines implies that CATms are

expected to conduct operations other than war particularly utilizing its different components to conduct DRRO.

c. **Highly Urbanized** – The continuous development of urban centers in the country makes urban areas inevitable battlefields in the future. Thus it is imperative for the Philippine Army to protect such urban areas and prepare for eventual engagement in urban terrain. The various vertical and horizontal constructions or their destroyed state in urban areas provide both advantages and disadvantages. Collapsed structures and other man-made obstacles can hamper both enemy and friendly forces' movement and the ability to immediately breach these obstacles will ensure the maneuverability and tactical advantage of CATms . On the other hand, protected areas such as cultural and heritage sites, hospitals, places of worship, and other places of refuge provide a ceiling in the optimal application of the destructive effects of Combined Arms Operations.

d. **Universally Connected** – With each passing day, the cyberspace domain is expanding its reach. Global economic, social, and communication systems are continuously being interconnected which makes the physical distance between people, organization, and other entities immaterial. This allows information to be disseminated through various platforms including the internet, telecommunications networks, and computer systems among others, at rates never seen before. This highly connected environment caused by the expansion of the cyberspace presents both opportunities and uncertainties for the Philippine Army. A connected and network centric CATms allows for the seamless vertical and horizontal execution of mission command between and among CATms and the operational headquarters. In addition, this helps provide the commander and other units a common operational picture of the operating environment which aides in decision making.

e. **Politically, Economically, and Socially Constrained** – The strategic direction of the Philippine Army and the AFP in general still anchors on the national priorities which are in turn influenced by the varying political and social landscape at the international and local level. The fluid nature of the international system and the country's economic development, cognizant of various economic pressures and fiscal limitations, eventually affects the Philippine Army's development. These constraints have direct effect on Combined Arms Operations in terms of developing the capabilities of the different combat arms for Combined Arms Operations.

f. **Collective Environment** – In general, collective environment is characterized by the collaboration of various government, non-government agencies, organizations, and partner nations towards the achievement of shared interests, concerns, responsibilities, and develop partnerships for the common good. For combined arms, this means that future operations, either military in nature or operations other than war will in one way or another be conducted in collaboration with other entities.

g. **Culturally Diverse** – The archipelagic nature of the Philippines and interaction with other countries paved the way for the development of different ethnic groups that have different cultural practices, norms, and traditions. Thus in order to establish legitimacy and generate popular support in military operations across the country, the idiosyncrasies of these different ethnic groups need to be considered.

The Landpower Maneuver

0116. Landpower Maneuver is the Philippine Army's land defense strategy that provides an overarching construct on how the Army, as part of the Joint Force, will conduct prompt and sustained land operations in order to secure the nation's sovereignty and land territory.

0117. The PA Landpower Maneuver calls for the organization of a Combined Arms Team (CATm) which shall be the basic organization of the Philippine Army combat force. The CATm is composed of the different Combat Arms, Combat Support and Combat Service Support units that are further categorized into Maneuver, Maneuver Support, and Combat Sustainment components to emphasize the required combined arms effect that the team must deliver. These components are likewise held together by effective Command and Control. The CATm has a Maneuver component to defeat a threat; a Maneuver Support component to ensure operational maneuverability; and a Combat Sustainment component to ensure force generation and regeneration throughout the operations. All in all, the execution of Combined Arms Operations ensures the accomplishment of four key tasks namely: defeat a threat; ensure operational maneuverability; gain popular support and legitimacy of action, and; support civil authorities.

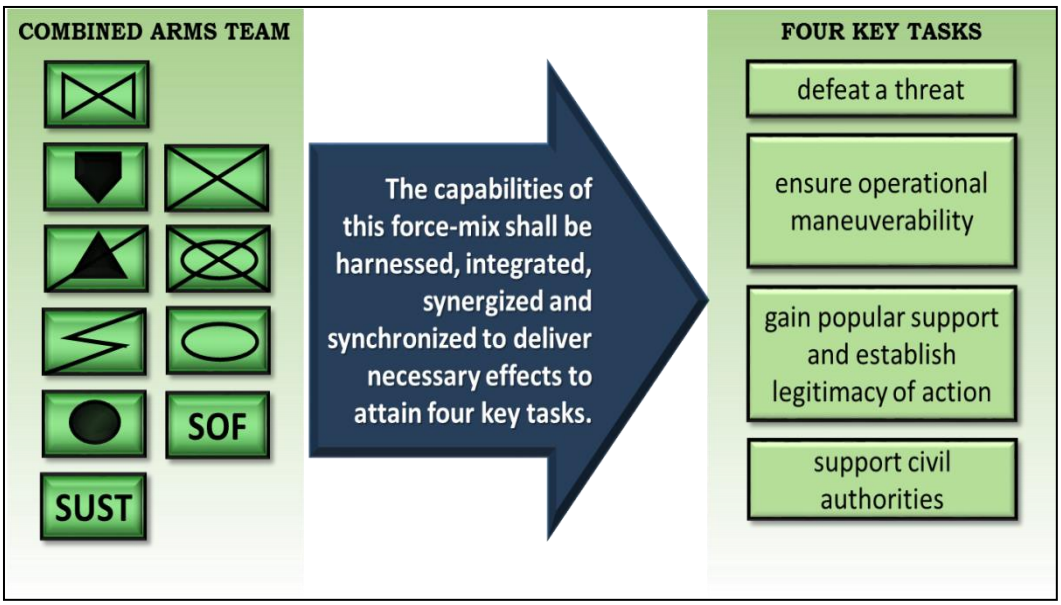


Figure 1.2 – Combined Arms Team in the Landpower Maneuver

0118. Combined Arms Operations is executed in both strategic and operational maneuvers. Strategic Maneuver encompasses the Army's posturing of its CATms. The strategic posturing of the Army's CATms will deter armed confrontation, and facilitate operational maneuvers to preempt, defeat, and destroy the threat as part of the overall defensive operations. During Operational Maneuver, Combined Arms Operations deliver the decisive effects in offensive, defensive, security, and support to civil authorities operations.

0119. The primary goal of Combined Arms Operations is the attainment of four key tasks namely to defeat a threat; ensure operational maneuverability; gain popular support and establish legitimacy of action; and support civil authorities. Defeating a threat refers to the condition. To ensure operational maneuverability, Combined Arms Operations defeats threats that allows for the execution of the different facets of Army operations particularly offense,

defense, security, and support to civil authorities. This includes activities that ensure the maneuverability of the force like mobility and countermobility tasks delivered by army combat engineers.

0120. Popular support and the establishment of legitimacy of action are vital elements to Combined Arms Operations. Popular support helps combined arms teams to harness the potential of the population to support army operations. Combined Arms Teams as part of the Philippine Army operates within the legal boundaries of both domestic and international laws. It adheres to the rule of law and the respects human rights and conventions on warfare. Combined Arms Operations support civil authorities by strengthening civil capacities in order to ensure stability.

Combined Arms Operations Framework

0121. The Combined Arms Operations Framework provides a logical way of understanding how Combined Arms Operations as an integral component of Landpower Maneuver contributes in addressing multiple challenges within the operational environment by employing all elements of Army power to achieve strategic and operational objectives. It also discussed the applicability of Combined Arms Operations as the Army's contribution to the joint force in the conduct of Full Spectrum Operations (offensive operations, defensive operation, security operations and support to civil authorities). More importantly, it illustrates how the different capabilities are integrated and synchronized to develop a proper force mixed, a combined arms team, and seamlessly executes a mission to achieve destructive, constructive, obstructive, or informational effects on the battlefield.

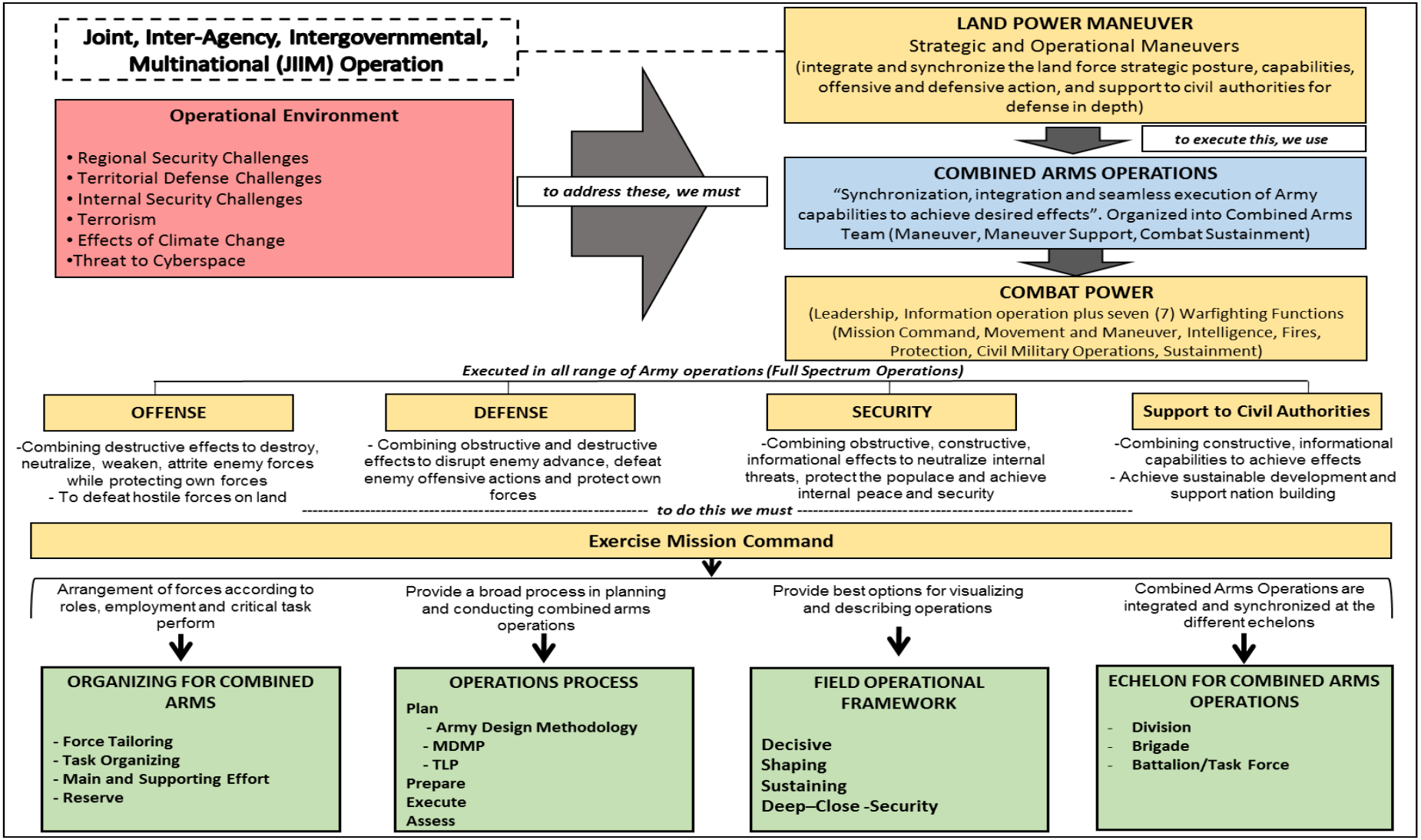


Figure 1.3. Combined Arms Operations Framework

0122. The emerging **operational and security environment** poses several threats and challenges that the Army will have to face in the future. This is coupled with the emerging notion of a **Joint, Inter-Agency, Intergovernmental and Multinational** operations wherein the Army will not exclusively operate within a particular physical environment but will in one way or another interact with other warm bodies or organizations. In this particular context, the Army formulated the **Landpower Maneuver** to provide the land defense strategy in conducting prompt and sustained land operations and its necessary elements in order to defeat the various threats and challenges. One of the major elements of Landpower Maneuver to defeat threats and challenges is through **Combined Arms Operations** or the synchronization, integration, and seamless execution of Army capabilities to achieve desired effects. The backbone of Combined Arms Operations is the synergy of Maneuver, Maneuver Support, and Combat Sustainment components.

0123. Maneuver, Maneuver Support, and Combat Sustainment emanates from the different warfighting functions under Army Power. **Army Power** is the summation of the potency that any total Army force generates. It is composed of the warfighting functions (Mission Command, Movement and Maneuver, Fires, Intelligence, CMO, Protection, and Sustainment) enhanced by information operation and held together by leadership. It is the total means of destructive, obstructive, constructive, and information capabilities that a miltiary unit can apply at a given time.

0124. Combined Arms Operations is executed across the Full Spectrum Operations that is from Offense, Defense, Security Operations, and Support to Civil Authorities Operations. Combined arms in **offensive operations** defeats threat on land by combining destructive effects to destroy, neutralize, weaken, and/or attrite enemy forces while protecting own forces. **Defensive operations** combines destructive and constructive effects to disrupt enemy advance, defeats enemy offensive actions, gains time for friendly forces, economize forces, and develops conditions favorable for offensive operations while protecting own forces. **Security operations** is combining obstructive, constructive, and informational effects to address internal security threats, protect the populace and achieve internal peace and security. **Support to civil authorities** combine constructive and informational capabilities in coordination with other instruments of national power to maintain or re-establish a safe and secure environment; provide essential government services, emergency infrastructure reconstruction, and humanitarian relief. Particularly, support to civil authorities operations strenghten the capacity of civil authorities for law enforcement, national development, natural resources and environmental protection, and disaster response and mitigaion efforts in order to ensure stability, which is critical to secure the country's strategic depth.

0125. Mission Command is the Army's preferred method of command and control to ensure the success of Combined Arms Operations across the Full Spectrum Operations. It is exercised through centralized planning and decentralized execution using mission orders. Mission Command is ensured through four (4) pillars namely Organizing for Combined Arms Operations, the Operations Process, the Field Operational Framework, and the Echelons for Combined Arms Operations.

a. **Organizing for Combined Arms Operations** is the arrangement of forces according to roles, employment, and critical tasks to be performed.

b. **The Operations Process** provides a broad process in planning and conducting Combined Arms Operations.

c. The **Field Operational Framework** provides the best options for visualizing and describing operations.

d. The **Echelons for Combined Arms Operations** decribe the different echelons in which Combined Arms Operations are integrated and synchronized.

0126. The succeeding chapters will further elaborate the different components of the Combined Arms Operations Framework.

Chapter II
Combined Arms Organization

0201. The Combined Arms Teams have Maneuver, Maneuver Support, and Combat Sustainment components. The purpose of Maneuver component is to defeat a threat and enhance civil capacity. It ensures operational and tactical advantage by positioning forces relevant to a threat. The purpose of Maneuver Support component is to ensure operational maneuverability. It enhances the effect that the maneuver component can deliver by providing fires and maneuver enhancement capabilities, mobility and counter-mobility, as well as ensure access and popular support. Combat Sustainment ensures the generation and regeneration of combat capabilities throughout the operations. While Command and Control is the exercise of authority and direction by a properly designated commander over assigned and attached forces in the accomplishment of a mission (PAM 3-00, Army Operations, 2011, p.94). It ensures the synergy of other key elements. Command gives authority to the commander. Command provides direction and leadership.

Combined Arms Operations integrates the employment of the different Maneuver, Maneuver Support, and Combat Sustainment capabilities in order to attain synergistic effects. The capabilities of the force mix shall be harnessed, integrated, synergized, and synchronized to deliver necessary effects.

Warfighting Functions

0202. In order to execute Combined Arms Operations, Army forces are organized into warfighting functions, each corresponding to an effect. A warfighting function is a group of tasks and related systems (people, organizations, information, and processes) united by a common purpose that commanders use to accomplish missions and training objectives *(ADRP 3-0, 2012, p.3-1).* These are:

a. Mission Command

b. Movement and Maneuver

c. Sustainment

d. Intelligence

e. Fires

f. Protection

g. Civil Military Operations

0203. Each warfighting function is subdivided into elements each with specific capability to achieve the function's overall effect. The elements are made up of organizations, equipment, facilities, personnel, and doctrines resulting in common purpose in terms of effect. Each element does not exclusively belong to a certain warfighting function and as such, elements

may also contribute to other warfighting functions. For example, Aviation platforms can be used to support movement and maneuver, provide ISR, and fires.

Warfighting Functions and Its Elements

0204. Figure 2.1 shows the components of a Combined Arms Team, the seven warfighting functions, and their elements.

Component	Warfighting Function	Elements	
Maneuver	Mission Command	Operational and Tactical Headquarters	
		C4S	
	Movement and Maneuver	Infantry	
		Armor	Tank
			Mech Inf
			Cavalry
		Special Operations Forces	
		Engineer	
		Aviation	
Maneuver Support	Fires	Artillery	
		Air Defense	
	Intelligence	Intelligence	
Combat Sustainment	CMO	CMO	
	Protection	Military Police	
		Explosive and Ordnance Disposal	
		Chemical, Biological, Radiological, and Nuclear	
		K-9	
	Sustainment	Logistics	
		Personnel Services	
		Health Services	

Figure 2.1 - Warfighting Functions and their Elements

Mission Command

0205. Mission Command as a warfighting function ensures the synergy of other key functions. It enables the commander the **art of command**, the creative and skillful exercise of authority through timely decision making and leadership, and the **science of control**, consisting of systems and procedures used to improve the commander's understanding and support accomplishing missions in order, to integrate the other warfighting functions.

0206. **Art of Command –** Command is the authority that a commander in the armed forces lawfully exercises over subordinates by virtue of rank or assignment. Commanders drive the operations process through understanding, visualizing, describing, directing, leading and assessing operations (Army design methodology).

0207. **Science of Control –** Control is the regulation of forces and warfighting functions to accomplish the mission in accordance with the commander's intent. Staffs support commanders in the exercise of mission command through the operations process – plan, prepare, execute, assess; conduct knowledge management and information management; and organizes as combined arms operations planning cell to synergize all efforts of the different warfighting functional staff.

0208. Operational and Tactical Headquarters – Operational and Tactical Headquarters provide commanders and their staff the physical space wherein activities related to the conduct of mission command are performed. Headquarters provides a facility for the commander to plan, prepare, control, and assess operations, issue orders, and coordinate with higher, lower, and adjacent units as well as synchronize the operation of lower echelon units. For CATms, it is important for each capability to have the proper representation in the staff or at the very least act as an SME or a liaison during the planning process to ensure that the utilization of the different capabilities is in accordance with its capacities and limitations.

0209. Command and Control, Communications, and Cyber Systems (C4S) – C4S provides the necessary vertical and horizontal means to communicate between combined arms teams and other units. It provides the means through which commanders through secured lines of communications like radio, intranet, and other means talk to each other, their subordinate units or Higher Headquarters. Having communication connectivity is vital for combined arms teams since proper coordination and synchronization is required to ensure efficiency in the employment of combined arms teams. C4S facilitates the tangible aspect of the Mission Command System through its personnel, networks, information systems, processes and procedures, facilities and equipment that enhances the commander's ability to conduct operations.

Movement and Maneuver

0210. The Movement and Maneuver warfighting function provides the decisive means by which the CATms engage the enemy. Movement and Maneuver function is to gain a position of advantage over an adversary through a combination of tactics and fires. Tactics is the employment and ordered arrangement of forces in relation to each other. Direct fire and Close Combat is inherent to maneuver. Movement is necessary to allow maneuver forces freedom of action in the operational area. Commanders use maneuver for massing effects of combat power to achieve surprise, shock, and momentum. This involves the following components:

a. **Infantry** – it has the ability to close in with the enemy through close combat in highly restricted (complex) terrain such as urban areas, mountains, and heavily vegetated areas. It enables army forces to dominate the land, either by destroying or capturing the enemy in difficult terrain or support civil agencies and protect populations and infrastructures.

b. **Armor** – can close in with the enemy through close combat by overwhelming combat power through mobility, protection, and firepower and conduct long range security and reconnaissance mission. Units under Armor are the tank, mechanized infantry, and cavalry. 1) Tank units can close with and neutralize enemy forces by fire, maneuver and shock effect in coordination with other units. It provides accurate, destructive fires that can be used to shape the battlefield and defeat any adversary. 2)

Mechanized Infantry can close with the enemy by means of fire and maneuver to destroy or capture the enemy or to repel enemy assault by fire, close combat, and counter attack. It complements tanks by providing a dismount capability which enables the unit to reduce obstacle and hold ground. The armored vehicle provides dismount team with mobility but less firepower and protection compared with the armored vehicles of tank units. 3) Cavalry performs reconnaissance, surveillance, and provides security for the unit to which it is assigned or attached. It can be employed in offensive, defensive or delaying action as part of the main battle or as an economy of force role. 4) Special Operations Forces (SOF) – provides a scalable option for crisis resolution, security, stability operations, and major confrontations. SOF provides unique and specialized capabilities to achieve military objectives of strategic and operational importance in hostile, denied, or politically sensitive areas.

c. **Combat Engineer** – provides the maneuver components freedom of action in the operational area. Its primary role is to provide mobility and counter-mobility operations. Mobility – provides engineering capabilities to commanders to gain and maintain a position of advantage against an enemy. Counter-mobility – denies the enemy freedom of action to attain a position of advantage against CATms.

d. **Aviation** – allows maneuver components rapid movement through vertical maneuver. It also provides close air support to maneuver units to inflict overwhelming firepower against the adversary.

Intelligence

0211. Intelligence is an important warfighting function for CATms. Intelligence function facilitates understanding of the enemy, terrain, and civil considerations. The purpose of intelligence is to ensure commanders, at all levels of command, have the critical capacity to make informed, accurate, and timely decisions. It includes tasks associated with support to situational understanding, targeting, information capabilities, force generation, and the conduct of ISR activities. Intelligence is more than just collection of information; it is a continuous process that analyzes information from all sources and conducts operations to develop the situation.

Fires

0212. Fires form part of the Maneuver Support component of the CATms. Fires support maneuver by destroying an adversary and providing wide area protection. It provides collective and coordinated use of indirect fires, air and missile defense, and joint fires through the targeting process. The following are its components:

a. **Field Artillery** – provides close fires and counter-fires capability to support maneuver; deep fires to extend fire capabilities; and interdiction fire to protect the force.

b. **Air Defense Artillery** - provides defensive capabilities to destroy aerial targets or missile, or reduce or nullify effectiveness of such attack either through surveillance actions or active engagements

Sustainment

0213. Sustainment provides support and services to ensure freedom of action, extend operational reach, and prolong endurance. Sustainment is the provision of logistics, personnel services, and health service support necessary to maintain operations until mission accomplishment. The Sustainment function ensures that CATms can operate in prolonged missions and not be routed due to the lack of ability to sustain the operations.

a. **Logistics** - is the planning and executing of the movement and support of forces. It includes those aspects of military operations that deal with: design and development; acquisition, storage, movement, distribution, maintenance, and disposition of materiel; acquisition or construction, maintenance, operation, and disposition of facilities; and acquisition or furnishing of services.

b. **Personnel Services** - complement logistics and health services by providing functions related to a soldier's welfare, readiness, and quality of life. In addition, it ensures the availability of adequate personnel to replace casualties in order to maintain the combat power of army units during operations.

c. **Health Services** - consists of all support and services performed, provided, and arranged by Army medical units in order to promote, improve, conserve, or restore the mental and physical well-being of soldiers. These include casualty care, medical evacuation, and medical logistics for army personnel.

Protection

0214. Protection intends to preserve the force so that the commanders of CATms can apply maximum combat power at a decisive point in order to accomplish the mission.

a. **Engineer Survivability** – capability to protect the force by altering the physical environment to provide or improve cover, concealment, and camouflage. Engineer Survivability can be done through construction of fighting positions, protective positions, hardened facilities, and camouflage and concealment.

b. **Chemical, Biological, Radiological and Nuclear (CBRN)** – protect the personnel, equipment, and property from Chemical, Biological, Radiological, and Nuclear threats. It conducts CBRN reconnaissance, and if hazards are found, conducts detection, identification and containment of CBRN hazards.

c. **Explosive Ordnance Disposal (EOD**) - provides ammunition management and disposal support to combined arms teams. It conducts render safe and disposal of unexploded ordnance, unserviceable ammunition, and improvised explosive devices.

d. **Military Police** – Military Police enforces military law and order, provide security, and custodial management in support to mission accomplishment. MP enforces military law and order, post rules and regulations, and maintains discipline inside army posts. In addition, MPs provide route and convoy security in order to facilitate the movement of troops and supply. Lastly, MPs provides custodial management to POWs, military detainees, and civilian internees.

e. **K-9** - provides protection for the force through the conduct of combat tracking operations, explosive detection operations, and scout patrol operations. In addition, K-9 can also be utilized in operations other than war particularly in search, rescue, and retrieval operations during HADR.

Civil Military Operations

0215. CMO include all actions taken to affect perceptions directly or indirectly (physical, non-physical, and non-lethal) to gain popular support, establish legitimacy, protect the force, populations, and influence the adversary's cognitive capability. It is characterized by activities that influence the beliefs, emotions, behaviors, attitudes, and opinions of intended audiences to facilitate military operations. At the tactical, operational, and strategic level, the CATms through CMO establishes and maintains good relations among its forces, government and non-government civil authorities, and the populace. CMO is characterized by its three pillars namely: Public Affairs, Civil Affairs, and Information Support Affairs.

a. **Public Affairs Pillar** conducts information-related activities directed to the external public to ensure clear, accurate, and timely dissemination of information to preserve security and privacy. Public Affairs is performed to establish and maintain constructive relations with the communities in order to promote the organization's objectives.

b. **Civil Affairs Pillar** are activities that promote public trust through cooperation, collaboration, and conduct of sociological and developmental activities to gain popular support. Civil Affairs are undertaken either independent of or coordination with civilian entities. In addition, Civil Affairs activities are designed to win the hearts and minds of the people; gain their willing cooperation, support, and confidence; and induce their active participation in the attainment of military objectives and national goals.

c. **Information Support Affairs Pillar** a CMO pillar aims to influence beliefs, emotions, attitudes, opinions, and behavior of the enemy, its mass base, and supporters to redirect unfavorable, neutral, or passive attitudes and behavior into favorable attitudes and behavior towards the friendly forces.

Combined Arms in Different Tactical Echelons

0216. Tactical echelons plan, prepare, and conduct Combined Arms Operations to achieve the combatant commander's objectives. CATms can be organized according to the following echelons:

a. **Division** - Division sized CATms are capable of planning operations and conducting maneuver of a force with three or four brigades, which may include allied brigades that can be integrated to the force. Moreover, division sized CATms has the highest operational staff and possesses all tactical-level warfighting functions. It can establish a Divisional Headquarters which acts as the Land Component Command under a joint headquarters.

b. **Brigade Combat Team** – A brigade sized CATm is the basic formation for Combined Arms Operations. It is responsible for the design and execution of contact

actions while coordinating combat support in its area of operations. It has limited internal capacity for tactical-level warfighting functions which is mitigated by the attachment of additional units to fill in these gaps. The optimal structure of a brigade sized combined arms team is composed of three to four maneuver battalions, maneuver support units, and sustainment units.

c. **Battalion Task Force** – Battalion Task Force level CATm conducts combined arms tactical maneuver. It is the core tactical element able to produce a significant effect in a mission assigned by a brigade. Battalions provide a specific service arms competency and always contain units from other warfighting functions. In terms of sustainment and the other warfighting functions, Battalion Task Forces have limited self-sufficiency. Battalions typically have four companies supported by maneuver support, and sustainment units.

d. **Special Operations Task Force (SOTF) -** Combined Arms Operations can also be done in situations where a Special Operations Area (SOA) is designated and other capabilities are required to accomplish a specific special operations task. In an instance where there is a specific special operations task, other components of the Army are placed under the tactical control of an SOTF for the duration of the mission.

Chapter III
Combined Arms in Full Spectrum Operations

0301. CATms conduct full spectrum operations in order to defeat a threat, ensure operational maneuverability, gain popular support and legitimacy of action and support civil authority. CATms execute full spectrum operations through offensive, defensive, and support to civil authority operations. It integrates the capabilities of the different arms to achieve destructive, obstructive, constructive, and informational effects to achieve military objectives. It performs different offensive, defensive, security, and support to civil authority tasks. However, the employment of tactics to accomplish the tasks may differ depending upon the land environment in which the CATms may operate.

Offensive Operations

0302. Offensive operations are combat operations to defeat and destroy enemy forces and seize terrain, resources, and population centers. CATms harness combat power to defeat, destroy, and neutralize enemy forces. Offensive action within the framework of land defense is designed to pre-empt, disrupt, diminish and destroy threats beyond the land domain, within a particular time. In addition, CATms conduct offensive operations to secure decisive terrain, deprive the enemy of resources, develop intelligence, deceive and divert enemy forces, hold enemy in position, disrupt the enemy's attack, and set up conditions for future successful operations.

Offensive Tasks

0303. There are four primary offensive tasks in conducting offensive operations:

a. Movement to Contact

b. Attack

c. Exploitation

d. Pursuit

0304. **Movement to contact**. A movement to contact develops the situation and establishes or regains contact. The objective is to enable CATms to make initial contact with a small element while retaining enough combat power to develop the situation and mitigate the associated risk. Movement to contact creates favorable conditions for subsequent tactical actions. Units conduct movement to contact when the enemy situation is vague or not specific enough to conduct the attack. A movement to contact result to a meeting engagement. On contact, the commander has five options: attack, defend, bypass, delay, or withdraw.

0305. The CATms is typically organized into movement to contact as a security force and the main body. Security forces are composed of advance guard or the covering force, the flank security and the rear security. The main body is composed of the remaining force. This may be composed of the main maneuver forces, maneuver support, and combat sustainment units.

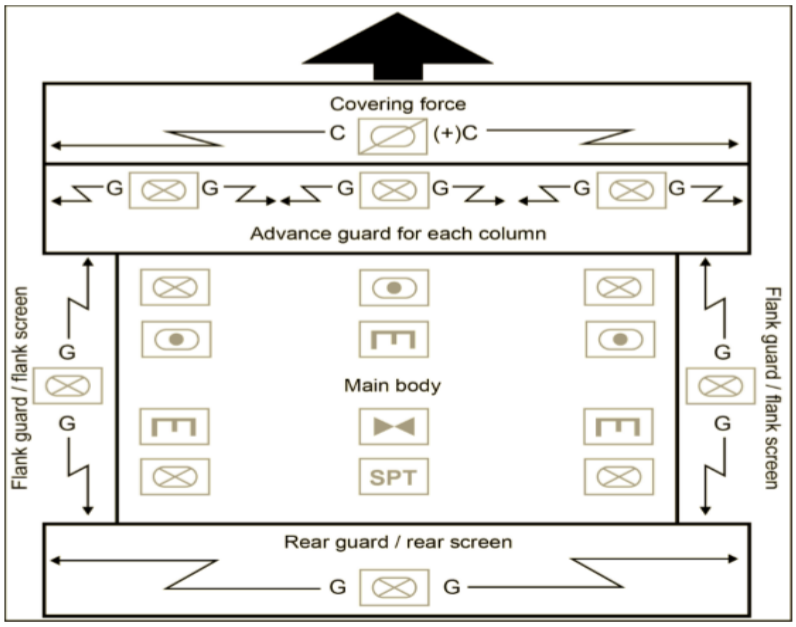


Figure 3.1 – Movement to Contact

0306. **Attack.** An attack destroys or defeat enemy forces, seizes and secures terrain, or both. Attacks require combined arms maneuver forces supported by its direct and indirect fires. It may be either decisive or shaping operations. Attacks maybe hasty or deliberate depending on the availability of time to plan and prepare. In the attack, CATms uses its intelligence assets to gather useable intelligence on the disposition and strength of the enemy, which allows the commander greater synchronization. This enables the commander mass combat power more effectively.

0307. During the attack, CATms organizes for Attack for the following:

a. Security forces

b. Main body

c. Reserve

d. Sustainment

0308. **Exploitation.** Exploitation rapidly follows a successful attack and disorganizes the enemy in depth. Exploitation is the bold continuation of an attack designed to increase success and take advantage of weakened or collapsed enemy defense. Exploitation capitalizes on a temporary advantage, on preventing the enemy from establishing an organized defense, or preventing the enemy from conducting an orderly withdrawal. Exploitation should prevent reconstitution of enemy defense, prevent enemy withdrawal, secure deep objectives, and destroy enemy command and control facilities, logistics, and forces. CATms conducts exploitation using their most mobile units to transform tactical success into a pursuit.

0309. **Pursuit.** A pursuit is designed to catch or cut off a hostile force attempting [1]to escape with the aim of destroying it. Pursuits occur when the enemy fails to organize a defense and attempts to disengage. If it becomes apparent that enemy resistance has broken down entirely and enemy forces are fleeing, a force can transition to a pursuit from any type of offensive or defensive operation. Pursuits require speed and decentralized control. Also, bold action, calculated initiative, and accounting for the associated risk are required in the conduct of a pursuit.

Sequence of offensive operation

0310. Offensive operations are conducted through phases whose rapid sequencing is critical. The following are the phases of offensive operations:

a. Gain and maintain enemy contact

b. Disrupt the enemy

c. Fix the enemy

d. Maneuver

e. Follow through

0311. This sequence illustrates the execution of offensive tasks that tends to overlap each other during the conduct of offensive actions. Normally the first three steps are shaping operations or supporting efforts, while the maneuver step is the decisive operation or main effort. Follow through is normally a sequel or a branch to the plan based on current operations.

a. **Gain and maintain enemy contact.** CATms gain and maintain enemy contact to identify enemy's composition, strength, and disposition. It provides the commander battlefield information that allows flexibility to commit forces to maneuver in a position of advantage. An advance guard or a covering force may initially make contact.

b. **Disrupt the enemy**. Prevent the enemy from organizing a defense or making a spoiling attack. The advance guard gathers as much information as possible and reports **about** the enemy's disposition, composition, strengths and weaknesses, capabilities and probable COA's.

c. **Fix the enemy.** The advance guard or lead element of the combined arms team fixes the enemy by preventing the enemy from maneuvering against the main body. It may **provide** the main body information about the enemy's disposition and strength. It may also provide a base of fire to allow the main body to maneuver and attack the enemy.

d. **Maneuver.** Once the enemy is fixed, the main body attacks enemy's weakness, either through a frontal, flank or rear while maintaining momentum of the advance. More

importantly, synchronization of maneuver and fires is critical to provide overwhelming combat power against the enemy.

e. Follow **through.** The Combined arms team transitions to continue the advance during **movement** to contact, pursuit and exploitation or termination of the operation once the enemy is defeated. Follow through may vary depending upon its assigned tasks.

Forms of Maneuver

0312. Forms of maneuver are tactical combinations of fire and movement employed by combined arms teams to maximize combat power against an enemy. In addition, forms of maneuver may vary depending upon the METTC. There are six forms of maneuver:

a. Envelopment

b. Turning Movement

c. Frontal Attack

d. Penetration

e. Infiltration

f. Flank attack

0313. **Envelopment.** Envelopment is a form of maneuver which a CATms seeks to attack the enemy's assailable flank. The force avoids enemy's strength which is usually in the front. Envelopment focuses *on* seizing terrain, destroy enemy's critical vulnerabilities by identifying specific forces in its flanks, or cutting off enemy's withdrawal routes.

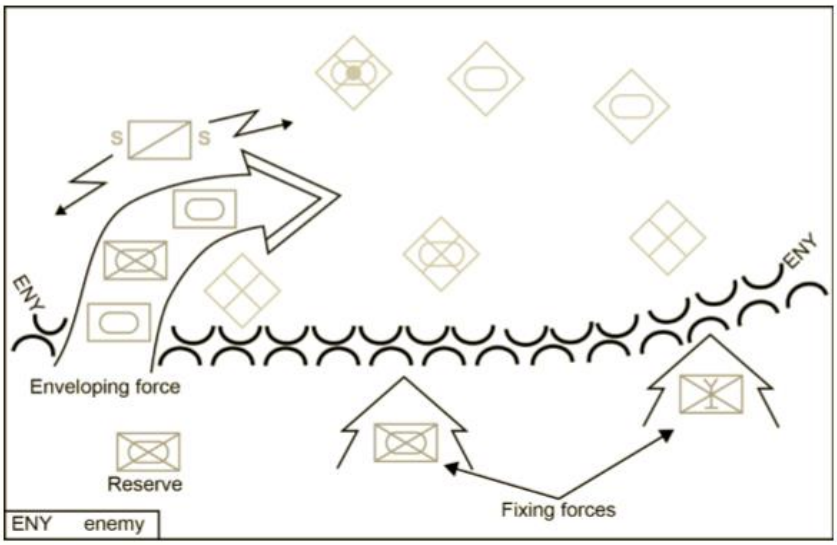


Figure 3.2 - Envelopment

0314. There are four varieties of envelopment:

a. Single envelopment

b. Double envelopment

c. Encirclement

d. Vertical envelopment

0315. The combined arms team using envelopment organizes its forces into two main functions: a fixing force and an enveloping force. However, it may designate its other forces in providing reconnaissance and securities, providing indirect fires to the enveloping or fixing force, breaching obstacles to allow enveloping force to penetrate an assailable flank, maintain a reserve for further exploitation or sustaining the envelopment operations.

0316. **Turning movement.** A turning movement is a form of maneuver which a CATms seeks to attack behind the enemy's current position in order to compel the enemy to move out into a position of disadvantage or divert its major forces to allow the CATms to shape the battle. Usually, the attacking force exploits the initiative by seizing enemy's vital lines of communication, support areas, or key terrain. Through turning movement, the enemy is forced to displace its current position and reposition, thus providing friendly forces more assailable sides for attack.

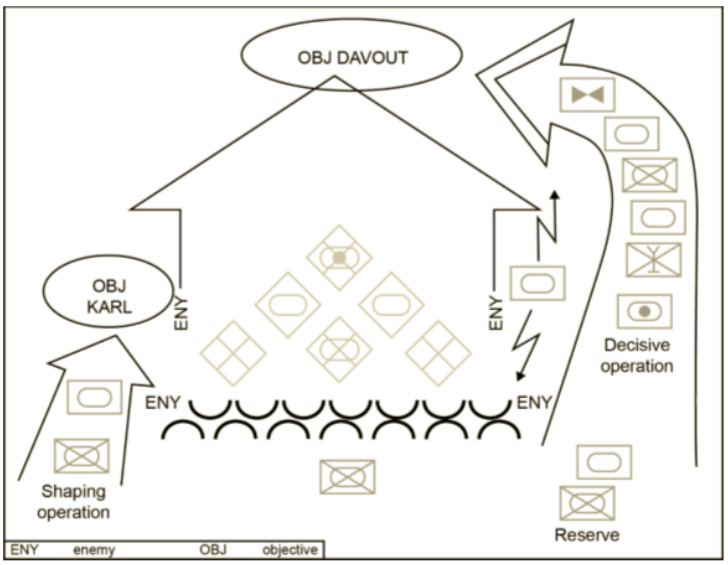


Figure 3.3 - Maneuver turning movement

0317. CATms organize into three main tasks: conduct a turning movement, conduct shaping operations, and conduct reserve operations. The turning force maneuver seizes vital terrain at the rear of the enemy's position. This will force the enemy to divert its forces and be dislocated from its current position. The turning force usually conducts its maneuver away from supporting distance from the main body. The turning force should at least contain maneuver support and sustainment capabilities to operate independently over a certain period of time. The commander of the CATms may designate the main body to conduct either

shaping operations or the decisive operations. The reserve also provides flexibility to the commander to either support the turning movement, shaping operations or counter enemy counteractions.

0318. **Infiltration.** An infiltration is a form of maneuver wherein small elements of the CATms covertly maneuver through enemy areas to occupy a position of advantage behind enemy positions. Infiltration is used in support to other forms of maneuver. SOF and light infantry are usually employed to avoid detection and engagement. In terms of infiltration, the CATms may designate more than one infiltration teams depending upon the situation.

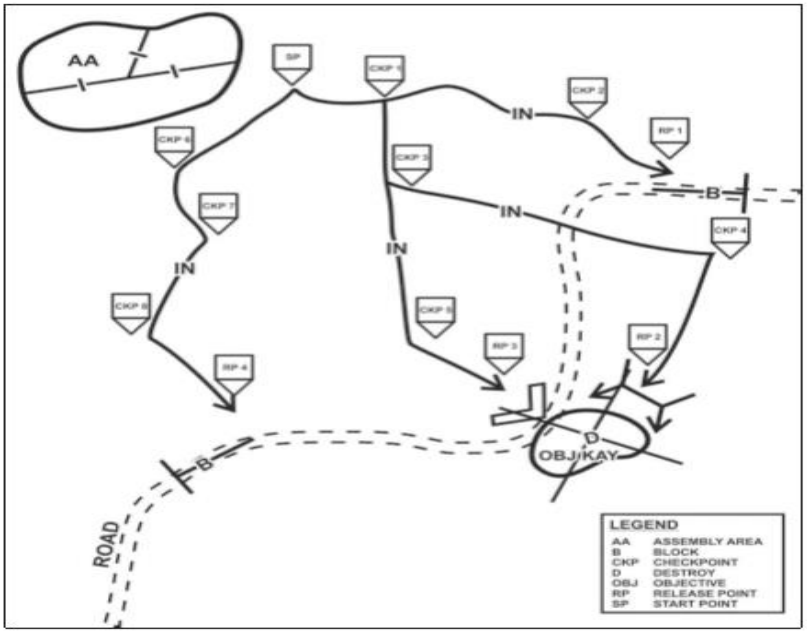


Figure.3.4 - Infiltration

0319. The Commander conducts infiltration in order to do reconnaissance on enemy positions, attack enemy positions in unexpected directions, seize key terrain, establish a base of fire in support to attacking elements, disrupt enemy lines of communications, command and control systems, and support areas, or conduct a undetected breach on enemy obstacles to allow follow-on forces to penetrate defensive positions.

0320. **Penetration.** A penetration is a form of maneuver in which CATms breaches enemy defense to seize the objective or destroy the enemy in depth. Penetration may cut across enemy security area through main defensive positions into enemy support areas. The commander organizes its forces to provide overwhelming combat power into the area of penetration to assure the breaching of enemy's defense and maintain the momentum of attack.

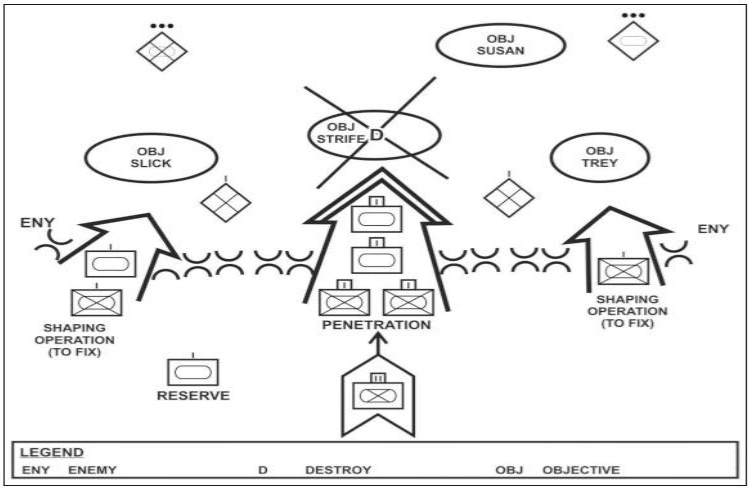


Figure 3.5 - Penetration

0321. The commander designates a breach, support, and assault force to do the penetration. Reserve forces provide flexibility in the penetration to foil enemy counterattack, act as follow-on force, or conduct exploitation.

0322. **Frontal attack.** When a CATm intends to destroy a weaker enemy force or fix a larger enemy force over a broad front, a frontal attack is employed. Frontal attack is a form of maneuver which is usually used to shape operations when fixing larger enemy forces in place to allow other forms of maneuver to take place. However, when the commander sees that the enemy is already in disarray, it uses frontal attack to exploit the situation and rapidly destroy the enemy.

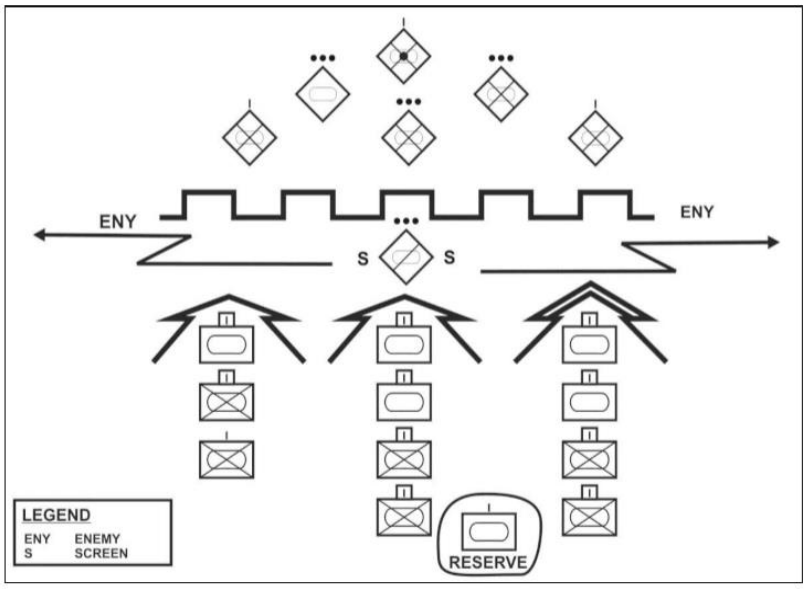


Figure.3.6 - Frontal Attack

0323. **Flank attack.** A flank attack is a form of maneuver when a CATms intends to defeat an enemy force in its flanks while minimizing the enemy's combat power usually oriented at the front. A flank may not be as strong in terms of forces or fires compared to a front of an enemy formation. Commanders of CATms use its main effort to attack at the flank while its supporting effort engages the enemy's front by maneuver and direct fires. Supporting effort also intends to divert the enemy's attention on its threatened flank.

Defensive Operation

0324. CATms conducts defensive operations to defeat an enemy attack, gain time, control key terrain, and economize forces. Defensive operations create conditions to transition to offensive operations to regain the initiative. As part of the land defense strategy, CATms conduct defensive operations to defeat an attack, facilitate a counter attack or preserve forces. This would enable forces to shift the initiative to destroy a threat or retake lost territories.

0325. There are three defensive tasks (operations):

a Area Defense

b Mobile Defense

c Retrograde

Area Defense

0326. These are defensive tasks (operations) conducted by the CATms in order to deny the enemy of key terrain, limiting their freedom of movement, and forcing them to move into a series of engagement areas where they are defeated, largely using mutually supporting direct and indirect fires. Area defense permit army forces retain defensive positions in depth. The commander of the combined arms team concentrates most of its combat power on designated defensive positions while retaining reserves either to reinforce defense positions or for counterattack.

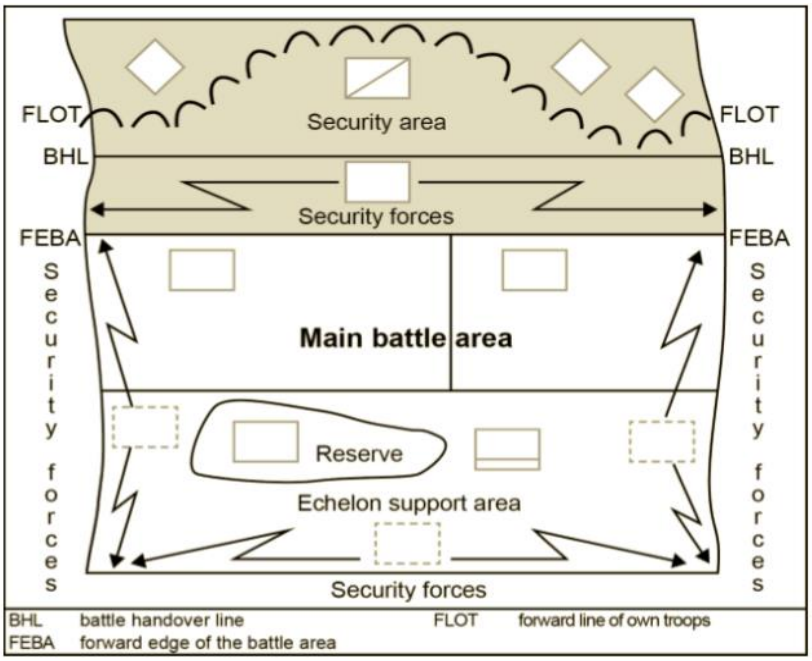


Figure 3.7 - Area Defense

Mobile Defense

0327. These are defensive tasks (operations) conducted by the CATms to concentrate its combat power to defeat an enemy force by means of counterattack. The defensive force allows the enemy to extend its forces in the depth of the area of operations, stretching its lines of communication, exposes its flanks, or over extends its combat power. The commander uses his CATms to shape the battle by fixing the enemy into position, channel its forces to engagement areas. He uses its decisive force to conduct a counterattack to destroy enemy main forces.

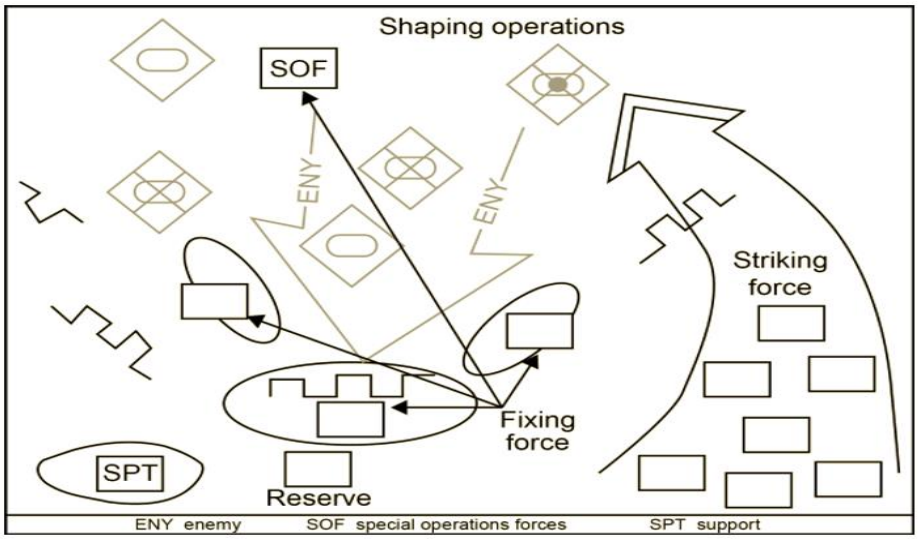


Figure 3.8 - Mobile Defense

Retrograde

0328. These are defensive tasks (operations) conducted by the CATms to move in an organized manner away from the enemy. Retrograde provides the commander options to preserve his forces, gain time, avoid combat under unfavorable situations, or place the enemy at a disadvantage position. Retrograde has three types: delay, withdrawal, and retirement.

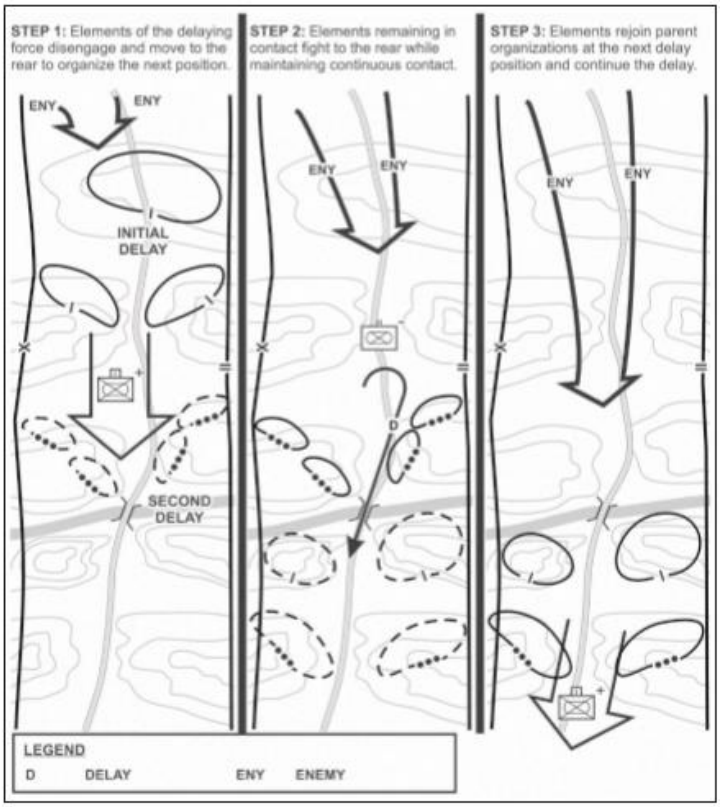


Figure 3.9 - Retrograde

0329. A **delay** is a retrograde operation wherein forces move rearward under pressure by the enemy in order to trade space for time by slowing down the enemy's momentum, inflicting significant damage, while not decisively engaged. CATms use its combat power to disrupt enemy advance in order to wear the adversary down, regain the initiative, and buy time to establish better defensive positions. A commander may use maneuver forces to alternately disengage against the enemy advance, employ engineer assets to put counter mobility obstacles, uses indirect fires to disrupt enemy advance, intelligence assets to provide ISR and sustainment units to protect lines of communications while conducting delay operations.

0330. CATms organize for delay operations by being divided into a main body, security force and a reserve. The main body conducts the delay operations by dividing it into two equal forces. The security forces provide a screen to the main body and at the same time provide information on enemy disposition, composition and strengths. The delaying forces either use alternate or successive positions. The reserve is used when the enemy attempts to penetrate its defense positions.

0331. A **withdrawal** is a retrograde operation wherein CATms disengages enemy contact and move away from enemy position. The commander conducts withdrawal to preserve his forces from enemy destruction. A withdrawal may or may not be under pressure. The CATms use all available combat power to ensure that the withdrawing forces be protected against enemy pursuit, exploitation, encirclement and destruction.

0332. A **retirement** is a retrograde operation in which a CATms does not engage or moves away from the enemy. The purpose of retirement is to avoid enemy contact in an undesirable

condition, disengages the force from battle to be used elsewhere, gain time, reposition forces, and secure more favorable terrain.

Forms of defense

0333. There are three (3) forms of defense:

a Defense of a linear obstacle

b Perimeter Defense

c Reverse Slope defense

0334. The commander of a CATm may conduct a defense along or behind a linear obstacle. Linear obstacles such as mountain ranges or river lines allow commanders to construct obstacles to stop enemy forces and channel them to planned engagement areas. CATms seek to defeat enemy attacking forces either through isolating them, conduct counterattacks, deliver fires on enemy concentration to defeat enemy attempts to breach the obstacle.

0335. A **perimeter defense** provides all around defense. When holding critical terrains such as holding a strong point, protecting a base, conducting a resupply, or conducting an operation in a noncontagious area of operations, a perimeter defense may be employed by a CATm. CATms present no assailable flanks to the enemy and allows the defender to reinforce a threatened area rapidly. Aggressive patrolling and security operations outside the perimeter are critical to a perimeter defense. Commanders may employ its forces forward along the perimeter or establish a defense in depth within the perimeter.

0336. **Reverse-slope defense** allows CATms to defeat enemy forces by using reverse-slope terrain that protects them from enemy's direct observation and fires. The reverse slope defense may deceive the enemy regarding its location and organization of its defense positions. The commander's goal is to make the enemy commit forces against the forward slope of the defense, causing the enemy to attack in uncoordinated fashion across the exposed topographical crest. Defending forces may surprise enemy forces on the crest, and engage them in massed fires. Commanders of CATms places over watching elements forward of the topographic crest and on the flanks of the position when organizing a system of reverse slope defense.

Security Operations

0337. Security Operations are combat operations to protect populations, forces and infrastructures and activities. Moreover, the operations conducted in coordination with other instruments of national power to maintain or re-establish a safe and secure environment and re-establish peace and order during crisis to conflict or normalcy after a resolution of a major crisis or war constitute security operations. During crisis to conflict, security operations contributes to securing land territory, the populace, and vital infrastructures against threats that undermine the Philippine political, economic, social systems, internal stability, national interests, and sovereignty and territory.

Counterinsurgency

0338. Counterinsurgency operations are military or paramilitary operations in coordination with other elements of national power to conduct, military political, economic, and psychological, and civic actions taken by the government to defeat an organized movement aimed at the overthrow of a constituted government through the use of subversion and armed conflict (PAM 3-00 Army Operations, 2014). The communist insurgency and the Moro secessionist movement are examples of threat groups that attempt to overthrow the Philippine government using politico-military means. In response, CATms uses all available capabilities to destroy the internal armed threat. Moreover, CATms conduct information operations and civil-military engagements to shape the operational environment in order to gain popular support, establish legitimacy of action and/or influence the adversary to leave their armed struggle.

0339. CATms organize for counterinsurgency operations by following the triad operational concept of integrating maneuver, intelligence and CMO. Examples include the Peace and Development Teams that conducts Bayanihan Team Activity in far flung areas to win the hearts and minds of the populace.

0340. Maneuver forces are supported by Maneuver Support and Combat Sustainment capabilities to perform their tasks. Also, artillery units provide indirect fire support. Intelligence units employ various intelligence disciplines to develop GESCON products to provide the commander a more enhanced understanding of the operational environment (Enemy and the Political, Economic, Socio-Cultural, Techno-Scientific). Engineer Units assists maneuver forces not only on enhanced mobility but reconstruction efforts in conflict-affected areas. Sustainment forces may assist either the maneuver or the maneuver support forces in providing logistics, personnel support, and health services to army forces and populations in conflict-affected areas.

Counter-terrorism

0341. Counter-terrorism are missions and tasks involving army forces to protect populations, installations and facilities, from terrorist acts initiated by foreign or local terrorists. Counter-terrorism is different from normal security operations because it entails long range planning and collaboration with other counter-terrorist agencies to identify, track, disrupt or destroy terrorist personalities, cells, and activities. In conducting counter-terrorism operations, SOF may be utilized for target interdiction and other special operations tasks that may be needed to accomplish the mission.

0342. CATms organize into smaller and highly mobile maneuver forces supported by maneuver support and sustainment units. Counter-terrorism also requires a robust intelligence network to destroy terrorist organizations operating in small and clandestine cells. To add, counter-terrorism also requires a joint, interagency and multinational collaboration. CATms or Army SOF may coordinate with other services to conduct counter terrorist operations depending upon the operating domain (maritime-land-air) especially in cases where service peculiar capabilities are needed to joint land operations. Army forces also conducts operations with other agencies such as the DILG-PNP, NBI, NICA and other agencies concerned in counter-terrorism. Also, CATms may conduct multilateral counter terrorist operations particularly against foreign-based terrorist cells like the Jemaah Islamiyah, ISIS, among others.

Support to Civil Security

0343. Other armed threat groups undermine the internal security of our country. CATms conduct support to civil security operations including the protection of vital installations and critical infrastructures against sabotage and armed threat. Furthermore, CATms can also be utilized in providing area security during high level engagements such the ASEAN Meeting and APEC Summit.

Support to Civil Authorities

0344. Support given to civil authorities encompasses military missions, tasks and activities in support to civil authorities to maintain or re-establish a safe and secure environment, provide essential government services, emergency infrastructure reconstruction, and human relief. In addition, support to civil authorities requires coordinated and integrated civil-military efforts.

Humanitarian Assistance and Disaster Response (HADR)

0345. HADR refers to missions and tasks conducted by army forces to respond to natural and/or man-made disasters in order to alleviate sufferings and prevent loss of lives and properties. Natural disasters like typhoons, earthquakes, volcanic eruptions and its resultant effects of flooding, landslides, storm surges, among others causes widespread human sufferings and extensive damage to properties and critical infrastructures. Man-made disasters are brought about or influenced by political, socio-economic and techno-scientific such as displacement of civilians in conflict-affected areas due to armed conflict. It may also be brought about by outbreak of deadly viruses that will cause widespread contamination that would claim loss of lives and/or chaos to the populace. The emergence of weapons of mass destruction may induce chemical, biological, radioactive and nuclear reactions that may also cause widespread loss of lives.

0346. The primary role of army forces during disaster response is to ensure security and stability in the affected areas in order to support local civil authorities, national line agencies, and other government and non-government organizations during the most critical periods within the emergency situations. CATms may conduct search, rescue and retrieval operations independently or in conjunction with other response units. CATms may also assist in relief operations, evacuation, distribution of emergency food and in providing temporary shelter and medical assistance.

0347. CATms may also perform other HADR tasks such as rehabilitation and recovery, disaster prevention and mitigation, and disaster preparedness. During rehabilitation and recovery, army forces assist civil authorities by restoring public utilities and reconstruct vital infrastructures. CATms in partnership with other stakeholders may take part in civic action activities such as tree planting, clean-up drives, disaster risk prevention interventions and other disaster prevention and mitigation activities. CATms may participate in disaster response drills; enhance HADR skills and equipment and training for disaster preparedness.

0348. CATms can be task-organized depending upon the requirements required during HADR. Maneuver units perform security operations to ensure stability during emergency situations. Other tasks include search, rescue, and retrieval (SSR) operations and relief

operations. K9 assets can be employed to Engineer assets will be employed during retrieval operations, rehabilitation and recovery by providing engineer assets in road clearing, repair and restoration of power and water supply. Command Posts shall be established with communication facilities to provide Command and Control capabilities to the lead agencies in HADR. Sustainment forces provide assistance in logistics, personnel support, and health services.

Support to Law Enforcement Operations

0349. When authorized and directed, CATms provide support to local and state law enforcement agencies in the implementation of penal and special laws. In extreme cases, when directed by the President, the Army maybe called upon to restore and maintain order (PAM 3-00, p 54). This may include performing tasks that is in support to law enforcement agencies such as counterdrug, peace and order, civil disturbance management operations, conduct of checkpoints, environmental protection and other law enforcement functions.

0350. Law enforcement support may be in the form of direct or indirect support. Direct support involves enforcing the law and engaging in physical contact with offenders. On the other hand, indirect support consists of aid to civilian law enforcement agencies but not enforcement of the law or direct contact with the offender (US ADRP 3-28 Defense Support to Civil Authorities, p 4-8). It is emphasized that law enforcement support shall be prescribed and regulated by legal authorities.

0351. CATms may be task organized as part of an interagency task force to perform support to law enforcement operations depending upon the support required by law enforcement agencies. It may provide command and control capabilities such as communications and cyber systems. Infantry and/or mechanized infantry forces provide additional law enforcement manpower. However, its employment shall be prescribed by legal authorities. When it comes to the collection of critical information, intelligence units through the different intelligence discipline may provide critical information to counterpart local and state authorities in its operational area. Law enforcement agencies may require specialized equipment, munitions and facilities provided by sustainment units to conduct its operations. Protection of critical facilities used during law enforcement operations can be provided by CATms to protect against a wide variety of hazards.

Support to National Development

0352. Support to National Development underscores the Army's commitment to lasting peace and security for the country. The Army conducts operations to establish conditions that prepare the communities for sustained development, provide military capabilities to capacitate and enable civil agencies to establish or restore basic services in communities, and provide them assistance in pursuing high-risk development projects.

0353. Army forces provide engineering and civic action activities to support local and national government in reconstructing and rehabilitating conflict-affected areas to create an environment conducive to further development. Engineering and civic action activities may include establishing or restoring basic infrastructures such as farm-to-market roads, school buildings, health centers, water systems and electrification projects.

0354. CATms form part of a joint task force to implement national development projects. Maneuver forces perform civic action tasks to identify national development projects in coordination with local government units. CATms also engage local agencies in formulating memorandum of agreements for the construction of identified projects. Engineer units construct roads, school buildings, and other national development projects while maneuver units perform security operations. Intelligence units provide situational understanding to the CATm performing national development tasks. Sustainment units provide logistics, health service support and personnel support.

Combined Arms Operations in Joint, Inter-Agency, Inter-governmental, and Multinational Operations (JIIM)

0355. Joint, Inter-Agency, Inter-governmental, and Multinational Operations (JIIM) provides an overarching construct on situations where a CATm operates as part of the joint force, or in collaboration with other national government agencies, local government entities or military forces from other nations.

0356. Joint operations employ two or more services in a single operation to achieve operational and strategic objectives. Joint operations perform in all range of military operations. The Army's contribution to joint operations is to provide combined arms land forces to dominate the land domain or support air or maritime domain. Likewise, the army organizes trains and equipment combined arms teams to contribute to the joint force commander in the conduct of joint operations. CATms shall integrate and synchronize with other services under the joint force through joint planning.

0357. In joint operations, Joint Force Commanders assign supporting or supported roles to the service components depending upon the type of joint operations being conducted. For land operations, the joint land defense forces are supported by air and maritime forces. In air operations, the supporting elements are the joint land and maritime forces. Lastly, for maritime operations, the supporting elements are the air and land forces.

Inter-Agency Cooperation

0358. Interagency operations are missions and tasks performed to uphold national sovereignty operations and interests, and support government. To expound, interagency cooperation is a form of operations where agencies, departments, and organizations, jointly perform a specific activity which aims to defend the country from external threat (AFP TD Manual s 2015, p 6-1). The AFP employs joint operations in a broader environment to include other instruments of national power. These should be operated in concert at all levels of warfare (from the Strategic to the tactical level). The key role of CATm is usually to create conditions for other instruments of national power to work. In times of crisis or war, CATm uses its combat power to defend its territory. In times of national calamities, land forces use its various capabilities to stabilize land areas for other government agencies to provide basic services. Similarly, it may assist organizations in relief and rehabilitation efforts using its maneuver support and sustainment assets. The CATm cooperates with other agencies in terms of law enforcement operations such as anti-illegal drugs operations and support to internally displaced persons and disaster relief operations.

Combined and Multinational Co-operations

0359. Combined Operations are operations conducted by forces of two or more nations, usually undertaken within the structure of a coalition or alliance. (AFP TD Manual s 2015, p 6-5). An alliance is a long-term relationship between nations resulting from a formal agreement with the objective of pursuing common interests of its members. A coalition is a short-term arrangement of two or more nations, commonly ad-hoc, performing activities for a common action.

0360. Land forces organize into combined arms teams to participate in combined operations, usually during multilateral exercises, peacekeeping operations, humanitarian assistance and disaster response. However, in times of war, commanders conduct Combined Arms Operations as part of a multinational force. The employment of military forces will be decided upon a political decision.

Combined Arms Operations in Specific Land Environments

0361. Specific land environments impose specific tactics, techniques, and procedures to CATms in the conduct of offensive, defensive, security, and support to civil authority tasks. Land environments have certain terrain and weather characteristics that may degrade or enhance the effectiveness in the employment of combined arms capabilities. Thus, it is crucial for commanders to understand and appreciate the land environment to maximize the employment of his combined arms capabilities.

Urban Areas

0362. Urban areas are characterized by complex man-made physical terrain superimpose on existing natural terrain. Compared to other land environments, urban areas have the highest population density. Its infrastructure is composed of super surface (such as roofs and upper floors of buildings), surface and sub-surface structures interconnected either randomly or in proper arrangement. (ATTP 3-06.11 Combined Arms Operations in Urban Terrain, p xii)

0363. Commanders use CATms that capitalize on adaptive and innovative leaders of small units (squads, platoons, companies). It may also employ SOF using its unique capabilities.

0364. Considerations on planning and conducting operations in urban terrain: (FT-04 Fundamentals of Combined Arms Operations, page 33)

a Technical considerations - observations and fields of fire, navigation on surface and sub-surface approaches, cover and concealment in reinforced structures.

b Tactical considerations – intermingling with units that may result to fratricide, protection for civilians and forces, close combat and fires techniques in heavily defensive structures.

c Psychological considerations – human consequences due to fatigue, casualties and collateral damage

Mountain Areas

0365. Mountain Areas are characterized by mountainous terrain, with heavy foliage, and rock formations. Also, the weather and climate in mountain areas may vary significantly depending on the time of the year and geographic location. It may vary significantly from wet and humid to dry and hot climate over a short period of time. It may affect troop's movement, health and even morale. Still, infantry elements of CATms can easily maneuver in these areas. On the other hand, the restrictions imposed by the terrain may hamper the maneuverability of large vehicles which can include armor units, sustainment convoys, and large artillery pieces.

0366. Enemy may capitalize on this environment using mastery of the terrain. To gain advantage against the enemy, commanders considers the following principles: (FT 0-4 Fundamentals of Combined Arms, p 33)

a Ubiquity – involves simultaneous and omnidirectional threat to the adversary in order to maximize surprise, disrupt his cohesion, and wear him down;

b domination – using high ground and vertical capabilities

0367. It is also important to consider the mobility of forces in this terrain, preservation of force from wear, sustainment of the CATms to increase its endurance, and the use of multiple intelligence disciplines.

Jungle Areas

0368. Jungle and mountainous terrain are somewhat similar environments. Jungle and mountainous terrain differ on the terrain feature since it usually rolling or flat terrain. Jungle areas have thick vegetation which slows down mobility, has limited view and fields of fire. It may be difficult to control units due to its thick foliage. The climate also poses challenge not only to the forces but also to the adversary. The road networks in these areas are sparse. It may be limited to roads and trails that are degraded by weather conditions.

0369. Jungle areas may also include thinly populated areas such as far-flung barangays and *sitios* which affect operations, particularly the rules of engagement and avoidance of human and collateral damage. Strict OPSEC procedures needs to be observed since some barangays are enemy-influenced that may compromise the security of army forces.

0370. Due to the restrictive nature of the terrain, commanders of CATm may rely on its infantry units in conducting operations in jungle areas. It may also employ mechanized-infantry if mobility corridors allow such capability. Successful operations in jungle areas combine dispersion and concentration. Small units such as recon teams or scout ranger teams are dispersed. Once the enemy makes contact, its other forces close in the enemy from all directions.

0371. Considerations in the conduct of operations in jungle areas include:

a Limited fields of view and fire;

b Difficulty in finding the enemy;

c Limited mobility

d Difficulty in controlling units.

Riverine Areas

0372. Riverine Areas pertain to littoral zones of coastal areas, inland waterways, marshland and its adjoining land areas. It is dominated by water lines of communication, which may consist of major rivers, tributaries, and an extensive network of waterways, canals, and irrigation ditches. Due to lack of suitable road net, movement of military units across riverine areas are through air and water. Consequently, suitable land bases, airfields, and artillery firing positions may not always be available in riverine areas.

0373. Historically, major settlements started in or near riverine areas. Riverine areas provide a logical place for the development and growth of population centers. Coastal waters and inland waterways are the major arteries for economic circulation as local settlers use it extensively as a conduit of various economic activities.

0374. Riverine areas can also be utilized by hostile forces particularly local insurgents, terrorist organizations, and other lawless elements for their activities. With this in mind, CATms conduct operations in riverine areas to deny its use to the enemy and ensure that riverine areas are open for economic activity as well as free from terroristic activities and untoward acts of lawless elements.

0375. Combined Arms operations forces conducts riverine operations to control waterways, conduct enemy interdiction, or infiltrate enemy held territory. CATms use water transport extensively to move troops and equipment throughout the area. Riverine operations forces involve infantry units and/or SOF maneuvers in watercraft formations in conducting riverine operations. Fire support must be pre-planned during water movements. Intelligence is critical along waterways. Sustaining riverine operations needs unique logistical requirements such as support boats and small crafts to transport personnel.

0376. Commanders plan and conduct riverine operations taking into consideration the following:

a The flat and absence of vegetation allows the operation to be easily detected by the enemy.

b Population centers are existent along the navigable or tributary waters.

c Weather is unpredictable and can change the river environment rapidly.

d Low water can change the channel depth below minimum requirements of watercrafts that impedes mobility.

e Difficulty in navigation dictates the need to establish control measures.

Chapter IV
Combined Arms Integration

0401. Commanders integrate combined arms to maximize army power in order to achieve its full destructive, obstructive, constructive and informational potential. A combined arm is the application of the different arms to achieve an effect greater than each arm was used separately or sequentially. It multiplies army forces effectiveness in all operations. Units operating without support of other capabilities generate less combat power and may not achieve its mission.

0402. Commanders of combined arms teams conceptualize and apply its army capabilities in terms of army power to accomplish its mission. Army power is composed of the leadership, information and the seven (7) warfighting functions namely mission command, movement and maneuver, fires, intelligence, sustainment, protection and civil military operations. Each warfighting function consists of related tasks and missions united by a common purpose that commanders use to accomplish the mission.

0403. The mission command warfighting function integrates and synchronizes the other warfighting functions to mass the effects of army power at the decisive place and time. Mission command alone does not accomplish missions and tasks. However, it provides the purpose and direction to the other warfighting functions.

0404. The framework for exercising mission command is the operations process – where major mission command activities are perform during operations. The Combined arms commander and staff use the operations process to integrate numerous tasks executed throughout the headquarters and with subordinate units.

0405. Commanders organize the headquarters into a tactical command center to perform planning, preparing, executing, and assessing operations. The staffs shall be reconfigured as an operational planning team with cells and elements clustered into warfighting functions. The commander ensures procedures to execute the operations process within the headquarters enable mission command. The staff uses standard operating procedures, battle rhythm, meetings, and running estimates to assist them in TOC (command post) operations.

Mission Command

0406. The Army's preferred method of exercising command and control is Mission Command (PAM 3-00 Army Operations, p 42). Mission Command is the conduct of military operations through centralized planning and decentralized execution using mission orders. Mission command allows subordinate commanders greatest possible freedom of action guided by the higher commander's intent using mission orders in order to seize, retain and exploit the initiative.

0407. Commanders, aided by the staff, exercise mission command by balancing the art of command and science of control in order to integrate the other warfighting functions. Commanders perform 3 primary mission command functions tasks: drive the operations process through understanding, visualizing, describing, directing, leading and assessing operations; develop teams; and inform and influence the audiences.

0408. The staff supports the commander and subordinate commanders with understanding situations, decision-making and implementing decisions through the conduct of operations.

0409. The following are the 4 staff tasks:

a Conduct the operations process

b Conduct knowledge management and information management

c Synchronize information related activities

d Conduct cyber electromagnetic activities

0410. The mission command system is composed of personnel, structures, and warfighting staff procedures that enable the combined arms commander to design the maneuver, develop orders, and plan, conduct and assess the action.

Combined Arms Operations Process

0411. Combined Arms Operations subscribes to the same operations process used throughout the Philippine Army. It provides a broad process in conducting operations, and consists of major command and control activities performed during operations namely: planning, preparing, executing, and continuously assessing the operation.

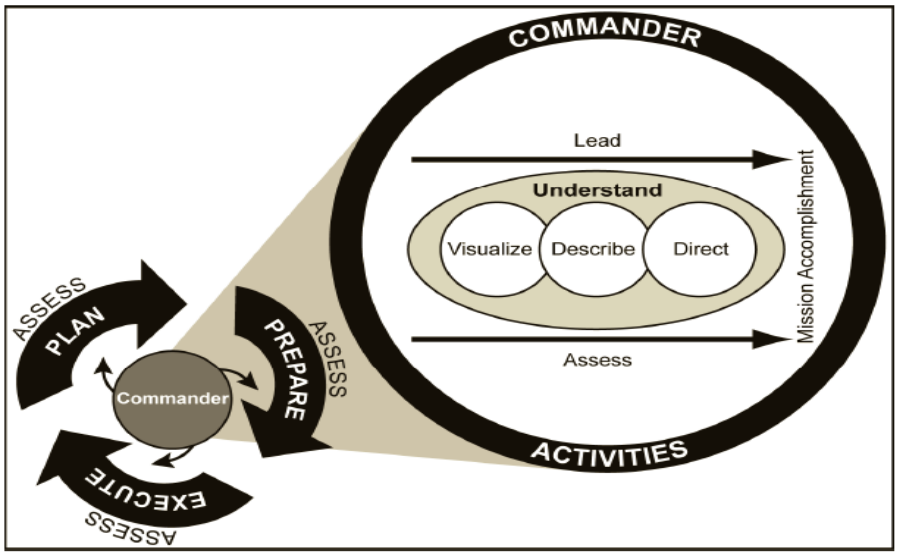


Figure 4.1 - Combined Arms Operations Process

0412. Upon receipt of a mission, planning starts a cycle of the operations process that results in a plan or operation order to guide the force during execution. After the completion of the initial order, however, the commander and staff revise the plan based on changing circumstances. While units and Soldiers always prepare for potential operations, preparing for a specific operation begins during planning and continues through execution. Execution puts plans into action. During execution, staffs focus on concerted action to seize and retain operational initiative, build and maintain momentum, and exploit success. As the unit executes the current operation, the commander and staff are planning future operations based on assessments of progress. Assessment is continuous and affects the other three activities.

Subordinate units of the same command may be conducting different operations process activities.

Role of Commander in the Operations Process

0413. Commanders are at the center of the operations process. They continuously develop, test, and update their understanding all throughout the conduct of operations. They are actively collaborating with other commanders and their staff, and other JIIM partners in order to create a shared understanding. As the commanders begin to develop an understanding of the operational environment, they start visualizing the operation's end-state and potential solutions to the solve problems. Then, they describe it to their staffs and subordinate for shared understanding. Commanders make decisions and direct the operations based on that deeper situational understanding. All throughout the operations, commanders lead soldiers and forces by providing direction and guidance. Continuous assessment on the situation allows commanders to better understand the progress of the operations and on how a dynamic operational environment affects it. Commanders incorporate the assessment of the staff, subordinate commanders, and other JIIM partners into their personal assessment of the situation. Based on the assessment, commanders modify plans and orders to better accomplish the mission.

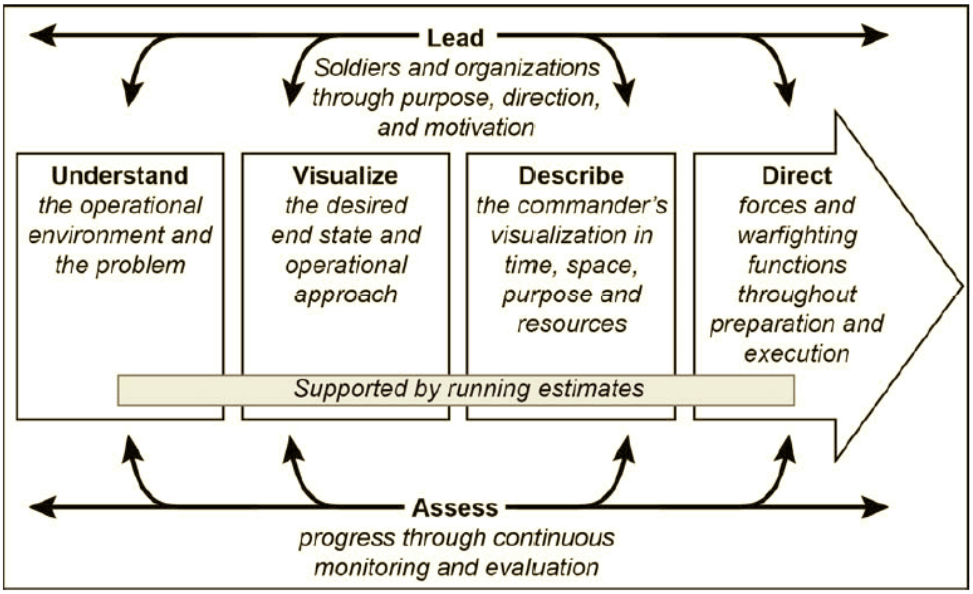


Figure 4.2 - The Role of the Commander in the Operations Process

Operational Art

0414. Operational Art is the cognitive approach by commanders and staffs – supported by their skill, knowledge, experience, creativity, and judgment – to develop strategies, campaigns, and operations to organize and employ military forces by integrating, ends, ways, and means. (ADRP 3-0 Unified Land Operations, p 4-1). For CATms, commanders aided by the staff use operational art to organize the different arms under their command through the arrangement of tactical actions in time, space, and purpose in order to achieve operational and strategic objectives. Through operational art, commanders translate their operational approach into a concept of operations and ultimately into tactical tasks.

0415. In applying operational art, the combined arms commanders and their staff uses the elements of operational art to help them understand the operational environment, as well as visualize, describe their approach for conducting their operations. These tools help commanders understand, visualize, and describe the integration and synchronization of the elements of combat power as well as their commander's intent and guidance. Commanders selectively use these tools in any operation. However, the tools' broadest application applies to long-term operations.

0416. The following are the elements of operational art:

a End-state and conditions

b Center of gravity

c Decisive points

d Lines of operations and lines of effort

e Operational reach

f Tempo

g Simultaneity and depth

h Phasing and transitions

i Culmination

j Risk

0417. The **end state** is a set of desired future conditions the commander wants to exist when an operation ends. Commanders include the end state in their planning guidance. A clearly defined end state promotes unity of effort; facilitates integration, synchronization, and disciplined initiative; and helps mitigate risk.

0418. A **center of gravity** is the source of power that provides moral or physical strength, freedom of action, or will to act. The center of gravity is a vital analytical tool for designing campaigns and major operations. It provides a focal point for them, identifying sources of strength and weakness.

0419. A **decisive point** is a geographic place, specific key event, critical factor, or function that, when acted upon, allows commanders to gain a marked advantage over an adversary or contribute materially to achieving success (JP 5-0). Decisive points help commanders select clear, conclusive, attainable objectives that directly contribute to achieving the end state. Geographic decisive points can include port facilities, distribution networks and nodes, and bases of operation. Specific events and elements of an enemy force may also be decisive points. Examples of such events include commitment of the enemy operational reserve and reopening a major oil refinery.

0420. **Lines of operations and lines of effort** link objectives to the end state. Commanders may describe an operation along lines of operations, lines of effort, or a combination of both. The combination of them may change based on the conditions within the area of operations. Commanders synchronize and sequence actions, deliberately creating complementary and reinforcing effects. The lines then converge on the well-defined, commonly understood end state outlined in the commander's intent.

0421. **Operational reach** is the distance and duration across which a joint force can successfully employ military capabilities (JP 3-0). It reflects the ability to achieve success through a well-conceived operational approach. Operational reach is a tether; it is a function of intelligence, protection, sustainment, endurance, and relative combat power. The limit of a unit's operational reach is its culminating point. It equalizes the natural tension between endurance, momentum, and protection.

0422. **Tempo** is the relative speed and rhythm of military operations over time with respect to the enemy or situation. It reflects the rate of military action. Controlling tempo helps commanders keep the initiative during combat operations or rapidly establish a sense of normalcy during humanitarian crises.

0423. **Simultaneity and depth** extend operations in time and space. Simultaneous combinations of offensive, defensive, and support to civil governance tasks overwhelm enemy forces and their will to resist while setting the conditions for a lasting, stable peace. Simultaneous actions across the depth of the operational area place more demands on enemy forces than they can effectively respond to them. Operations combining depth and simultaneity achieve a synergy that paralyzes enemy forces.

0424. A **phase** is a planning and execution tool used to divide an operation in duration or activity. A change in phase usually involves a change of mission, task organization, or rules of engagement. Phasing helps in planning and controlling and may be indicated by time, distance, terrain, or an event. The ability of Army forces to extend operations in time and space, coupled with a desire to dictate tempo, often presents commanders with more objectives and decisive points than the force can engage simultaneously. This may require commanders and staffs to consider sequencing operations.

0425. **Transitions** mark a change of focus between phases or between the ongoing operation and execution of a branch or sequel. Shifting priorities between the core competencies or among offensive, defensive, stability, and defense support of civil authorities tasks also involve a transition. Transitions require planning and preparation well before their execution to maintain the momentum and tempo of operations. The force is vulnerable during transitions, and commanders establish clear conditions for their execution.

0426. The **culminating point** is that point in time and space at which a force no longer possesses the capability to continue its current form of operations. Culmination represents a crucial shift in relative combat power. It is relevant to both attackers and defenders at each level of war. While conducting offensive tasks, the culminating point occurs when the force cannot continue the attack and must assume a defensive posture or execute an operational pause. While conducting defensive tasks, it occurs when the force can no longer defend itself and must withdraw or risk destruction. The culminating point is more difficult to identify when Army forces conduct stability tasks. Two conditions can result in culmination: units being too dispersed to achieve wide area security and units lacking required resources to

achieve the end state. While conducting defense support of civil authority's tasks, culmination may occur if forces must respond to more catastrophic events than they can manage simultaneously. That situation results in culmination due to exhaustion.

0427. **Risk**, uncertainty, and chance are inherent in all military operations. When commanders accept risk, they create opportunities to seize, retain, and exploit the initiative and achieve decisive results. The willingness to incur risk is often the key to exposing enemy weaknesses that the enemy considers beyond friendly reach. Understanding risk requires assessments coupled with boldness and imagination. Successful commanders assess and mitigate risk continuously throughout the operations process.

Planning

0428. Planning is the 1st phase in the operations process. A plan is a design for a future or anticipated operations and is a continuous evolving framework of anticipated actions that guides subordinates through each phase of the operation. Plans come in many forms and vary in scope, complexity, and length of planning horizons. Planning is the process by which commanders of combined arms teams and staff translate the commander's visualization into a specific course of action for preparation and execution, focusing on expected results. It involves an analysis and assessment of the conditions in the operational environment, an understanding and framing of the problem, and envisioning the set of conditions that represent the desired end state.

0429. Planning consists of two separate, but closely related, components: a conceptual component and a detailed component. Conceptual planning involves understanding the operational environment and the problem, determining the operation's end state, and visualizing an operational approach. Conceptual planning generally corresponds to operational art and is the focus of the commander with staff support. Detailed planning translates the broad operational approach into a complete and practical plan. Generally, detailed planning is associated with the science of operations including the synchronization of the forces in time, space, and purpose. Detailed planning works out the scheduling, coordination, or technical problems involved with moving, sustaining, and synchronizing the actions of force as a whole toward a common goal. Effective planning requires the integration of both the conceptual and detailed components of planning.

Army Planning Methodologies

0430. Army leaders employ three methodologies for planning. Commanders and staffs determine the appropriate mix of these methodologies based on the scope of the problem, their familiarity with it, the time available, and the availability of a staff. Methodologies that assist commanders and staffs with planning include the following: (ADP 5-0 Operations Process, page 7)

a Army Design Methodology

b Military Decision Making Process (MDMP)

c Troop Leading Procedures (TLP)

0431. **Army Design Methodology** is a method of applying critical and creative thinking to understand, visualize, and describe problems and approaches to solving them. The commander and staff conduct conceptual planning to understand, visualize, and describe the operational environment and the operational approach. Outputs of the conceptual planning include a problem statement, draft mission statement, draft commander's intent, a broad concept sketch, initial decision points, commander's critical information requirements, and initial planning guidance. This output will be used to begin detailed planning.

0432. The **Military Decision Making Process** provides a structure to help commanders and staff develop estimates, plans, and orders. It is the preferred method of planning for combined arms operations since it involved the integration of the different warfighting functions. It provides a logical sequence for decision making and interaction between commander and staff, and provides a common framework for parallel planning. Shown on figure 4.5 is the framework on the MDMP process.

Key Inputs	Steps	Key Outputs
• Higher headquarters' plan or order or a new mission anticipated by the commander	**Step 1: Receipt of Mission**	• Commander's initial guidance • Initial allocation of time
	Warning Order	
• Higher headquarters' plan or order • Higher headquarters' knowledge and intelligence products • Knowledge products from other organizations • Army design methodology products	**Step 2: Mission Analysis**	• Problem statement • Mission statement • Initial commander's intent • Initial planning guidance • Initial CCIRs and EEFIs • Updated IPB and running estimates • Assumptions
	Warning Order	
• Mission statement • Initial commander's intent, planning guidance, CCIRs, and EEFIs • Updated IPB and running estimates • Assumptions	**Step 3: Course of Action (COA) Development**	• COA statements and sketches -Tentative task organization -Broad concept of operations • Revised planning guidance • Updated assumptions
• Updated running estimates • Revised planning guidance • COA statements and sketches • Updated assumptions	**Step 4: COA Analysis (War Game)**	• Refined COAs • Potential decision points • War-game results • Initial assessment measures • Updated assumptions
• Updated running estimates • Refined COAs • Evaluation criteria • War-game results • Updated assumptions	**Step 5: COA Comparison**	• Evaluated COAs • Recommended COAs • Updated running estimates • Updated assumptions
• Updated running estimates • Evaluated COAs • Recommended COA • Updated assumptions	**Step 6: COA Approval**	• Commander-selected COA and any modifications • Refined commander's intent, CCIRs, and EEFIs • Updated assumptions
	Warning Order	
• Commander-selected COA with any modifications • Refined commander's intent, CCIRs, and EEFIs • Updated assumptions	**Step 7: Orders Production, Dissemination, and Transition**	• Approved operation plan or order • Subordinates understand the plan or order

CCIR – Commander's Critical Information Requirement
COA – Course of Action
EEFI – Essential Elements of Friendly Information
IPB – Intelligence Preparation of the battlefield

Figure 4.3 - The MDMP Framework

0433. At lower tactical echelons, commanders and leaders follow **Troop Leading Procedures**. Troop leading procedures are a dynamic process used by small-unit leaders to analyze a mission, develop a plan, and prepare for an operation (ADP 5-0). MDMP and Troop Leading Procedures are means to an end and their inherent value lies in the results

achieved, as such, they hinge on the commander's ability to visualize and describe the operation.

Components of a Plan

0434. The unit's task organization, mission statement, commander's intent, concept of operations, tasks to subordinate units, coordinating instructions, and control measures are key components of a plan. Commanders ensure their mission and end state nest with those of their higher headquarters. While the commander's intent focuses on the end state, the concept of operations focuses on the way or sequence of actions by which the force will achieve the end state. The concept of operations expands on the mission statement and commander's intent. Within the concept of operations, commanders may establish objectives as intermediate goals toward achieving the operation's end state. When developing tasks for subordinate units, commanders ensure that the purpose of each task nests with the accomplishment of another task, with the achievement of an objective, or directly to the attainment of an end state condition.

0435. **Task organization** is a temporary grouping of forces designed to accomplish a particular mission**.** Commanders task organize the force by establishing command and support relationships. Command relationships define command responsibility and authority. Support relationships define the purpose, scope, and effect desired when one capability supports another. The unit's task organization is in the base plan or order or in Annex A (Task Organization).

0436. The **mission statement** is the task, together with the purpose, that clearly indicates the action to be taken and the reason therefore (JP 3-0). Commanders analyze a mission as the commander's intent two echelons above them, specified tasks, and implied tasks. They also consider the mission of adjacent units to understand how they contribute to the decisive operation of their higher headquarters.

0437. The **commander's intent** succinctly describes what constitutes success for the operation. It includes the operation's purpose, key tasks, and the conditions that define the end state. It links the mission, concept of operations, and tasks to subordinate units. A clear commander's intent facilitates a shared understanding and focuses on the overall conditions that represent

0438. The **concept of operations** is a statement that directs the manner in which subordinate units cooperate to accomplish the mission and establishes the sequence of actions the force will use to achieve the end state. The concept of operations expands on the commander's intent by describing how the commander wants the force to accomplish the mission. It states the principal tasks required, the responsible subordinate units, and how the principal tasks complement one another. Commanders and staff use the following operational frameworks to help conceptualize and describe their concept of operation:

a The deep-close-security framework to describe the operation in time and space

b The decisive-shaping-sustaining framework to articulate the operation in terms of purpose

c The main and supporting effort framework to designate the shifting prioritization of resources

0439. **Tasks to subordinate units** direct individual units to perform a specific action. Tasks are specific activities that contribute to accomplishing missions or other requirements. Tasks direct friendly action. The purpose of each task should nest with completing another task, achieving an objective, or attaining an end state condition.

0440. **Coordinating instruction**s apply to two or more units. They are located in the coordinating instructions subparagraph of paragraph 3 (execution) of plans or orders. Examples include CCIRs, fire support coordination and airspace coordinating measures, rules of engagement, risk mitigation measures, and the time or condition when the operation order becomes effective.

0441. Commanders exercise control through control measures established throughout an OPLAN or OPORD. A *control measure* is a means of regulating forces or warfighting functions (ADP 6-0). Control measures are established under a commander's authority; however, commanders may authorize staff officers and subordinate leaders to establish them. Commanders may use control measures for several purposes: for example, to assign responsibilities, require synchronization between forces, impose restrictions, or establish guidelines to regulate freedom of action. Control measures are essential to coordinating subordinates' actions. They can be permissive or restrictive. Permissive control measures allow specific actions to occur; restrictive control measures limit the conduct of certain actions.

Organizing for Combined Arms Operations

0442. Combined Arms Operations are conducted through organizational design and temporary reorganization. When required, these capabilities are added through force tailoring and task organizing different types of brigades and battalions under a division headquarters. The task organized units further specifies the designation of a main effort, one or several units as supporting effort, and a formidable reserve.

0443. **Force tailoring** is the process of determining the right force mix of units and the sequence of their deployment in support of a joint force commander or a CATm commander. It involves selecting the right force structure for a joint or a Combined Arms Operations from available units within a combatant command or from the Army force pool; in other words, it arrays all available forces that can be utilized for a Combined Arms Operations. For CATms, force tailoring is influenced by the physical and non-physical characteristics of the operating environment. For example, terrain considerations may limit the applicability of armor units to a particular operation in a similar vein; adverse weather conditions can affect the deployment of army aviation units in support of an operation.

0444. **Task organizing** is the art of designing an operating force, support staff, or logistic package of specific size and composition to meet a unique task or mission. Characteristics to examine when task organizing the force includes, but are not limited to: training, experience, equipage, sustainability, operating environment, threat, and mobility. In CATms, it includes allocating available assets to subordinate commanders and establishing their command and support relationships. These will be further discussed on the following section.

0445. The **main effort** is the unit, battle positions, sector, zone, axis, avenue of approach, area of operations; theater of operations, and so forth, the senior commander has determined has the most important task and purpose at that time, whose success will make the most difference in the accomplishment of the higher commander's overall mission or objective. The **supporting effort** supports the success of the main effort. It directly or indirectly, creates a necessary condition for the main effort to succeed. **Reserves** are a portion of a force withheld from action or uncommitted to a specific course of action, so as to be available for commitment at the decisive moment. Its primary purpose is to retain flexibility through offensive action.

Army Command and Support Relationship

0446. Commanders of CATms task organize by establishing command and support relationship. Command relationships define command responsibility and authority. Support relationships define the purpose, scope, and effect desired when one capability supports another. The unit's task organization is in the base plan or order or in Annex A (Task Organization).

0447. Establishing a defined command and support relationship is essential in organizing CATms for operations. Following the concept of operations, these relationships establish clear lines of responsibilities and authorities between subordinate and supporting units. This allows the commander to effectively organize his forces and helps supporting commanders to understand their unit's role in the organizational structure.

Army Command Relationships

0448. Command relationships define superior and subordinate relationships between unit commanders. By specifying a chain of command, command relationships unify effort and enable commanders to use subordinate forces with maximum flexibility. Army command relationships identify the degree of control of the gaining Army commander. The type of command relationship often relates to the expected longevity of the relationship between the headquarters involved and quickly identifies the degree of support that the gaining and losing Army commanders provide. Army command relationships include organic, assigned, attached, operational control (OPCON), tactical control (TACON), and administrative control (ADCON).

0449. **Organic forces** are those assigned to and forming an essential part of a military organization. Organic parts of a unit are those prescribed in the Table of Organization and Equipment (TOE) for the Philippine Army. If temporarily task-organized with another headquarters, organic units return to the control of their organic headquarters after completing the mission. For example, a Field Artillery battalion is organic to the Army Artillery Regiment.

0450. **Assign** is to place units or personnel in an organization where such placement is relatively permanent, and/or where such organization controls and administers the units or personnel for the primary function, or greater portion of the functions, of the unit or personnel.

0451. **Attach** is the placement of units or personnel in an organization where such placement is relatively temporary (JP 3-0). A unit that is temporarily placed into an organization is an attached unit.

0452. **Operational Control (OPCON)** is the transferable command authority that may be exercised by commanders at any echelon at or below the level of combatant command. Operational control is inherent in a combatant command. Operational control may be delegated and is the authority to perform those functions of command over subordinate forces involving organizing and employing commands and forces, assigning tasks, designating objectives and giving authoritative direction necessary to accomplish a mission. Operational control includes authoritative direction over all aspects of military operations and joint training necessary to accomplish missions assigned to the command. Operational control should be exercised through the commanders of subordinate organizations. Normally this authority is exercised thorough subordinate joint force commanders and service and/or functional component commanders. Operational control normally provides full authority to organize commands and forces and to employ those forces, as the commander in operational control considers necessary to accomplish assigned missions. Operational control does not, in itself, include authoritative direction for logistics or matters of administration, discipline, internal organization, or unit training. In essence, when subordinate units are under operational control, commanders may assign missions and tasks, deploy units, and direct forces for its inherent mission/task. However, commanders cannot reassign forces and assign separate employment of components. (PAM 3-00, page 29-30)

0453. **Tactical Control (TACON)** is the command authority over assigned or attached forces or commands, or military capability or forces made available for tasking, that is limited to the detailed and, usually, local direction and control of movements or maneuvers necessary to accomplish missions or tasks assigned. Tactical control is inherent in operational control. When subordinate units are under tactical control, commanders may deploy units and direct forces for its inherent mission/task. However, commanders cannot assign missions and tasks, reassign forces and assign separate employment of components. (PAM 3-00, page 30)

0454. **Administrative Control (ADCON)** is the direction or exercise of authority over subordinate or other organizations with respect to administration and support, as maybe specified by the delegating authority. ADCON may be delegated to and exercised by commands of major services forces at any echelon at or below the level of service component command.

Army support relationship

0455. Army Support Relationships guide the action of a combat support or combat service support unit that aids, protects, complements, or sustains another unit. Army Support relationships allow supporting commanders to employ their units' capabilities to achieve results required by supported commanders. A supporting unit remains under the command of its parent headquarters. However, the supporting unit is authorized and required, within the context of the directive establishing the relationship, to respond directly to the supported unit's request for support. Army support relationships are direct support (DS), general support (GS), reinforcing (R), and general support-reinforcing (GSR).

0456. **General Support (GS)** provides support to the supported force as a whole and not to a particular subdivision. A subdivision cannot directly request support from a GS unit. Only supported force HQs determines the priorities and assigns the mission or task. It has no command relationship with the supported unit or force.

0457. **Direct Support (DS)** provides support to a specific unit or force. The supporting unit takes support requests directly from the supported unit or force and provides advice to the supported unit as to the use of the former's capabilities. The unit in direct support has no command relationship with the supported unit or force.

0458. **Reinforcing (R)** provides support to a similar supporting unit. Only like units (for example, artillery to artillery) can be given a reinforcing mission. The reinforcing unit has no command relationship with the supported unit.

0459. **General Support Reinforcing (GSR)** is a supporting unit that provides support to a force as a whole and provides reinforcing support to a similar supporting unit secondarily. It has no command relationship with the supported units or forces.

Field Operational Framework

0460. The Field Operational Framework provides best options for visualizing and describing operations in terms of time, space, purpose, resources, enemy, and the situation. It is a cognitive tool used by commanders and staff to conceptualize and describe the concept of operations. Commanders of combined arms teams may utilize one of three conceptual frameworks namely:

a Deep-close-security operations framework to describe operations in time and space;

b Decisive-shaping-sustaining framework to articulate operations in terms of purpose, and;

c Main and support efforts framework to designate shifting prioritization of resources.

0461. The **deep-close-security** operations framework has always been associated with terrain orientation but has also been applicable to temporal and organizational orientations as well. Deep operations involve efforts to prevent uncommitted enemy force from being committed in a coherent manner. This involves the utilization of deep fires assets of combined arms teams to prevent the enemy from employing long-range cannon, rocket, or missile fires. It may also involve activities by SOF units that disrupt enemy activities. Close operations on the other hand are operations against a hostile force in immediate contact and are often decisive operations. Security operations involve efforts to provide an early and accurate warning of enemy operations and to provide time and maneuver space within which to react to the enemy.

0462. **Decisive-shaping-sustaining** operations framework is a broad conceptual framework that includes activities that accomplishes the mission, shapes the environment, and sustains combat power. A decisive operation is the operation that directly accomplishes the mission. It determines the outcome of a major operation, battle, or engagement wherein a combined arms team or multiple combined arms teams engage hostile forces to accomplish the mission. Shaping operations establishes the conditions for the decisive operation through effects on the

enemy, other actors, and the terrain. Sustaining operations are focused on friendly forces in order to sustain decisive and shaping operations by generating and maintaining the combat power of combined arms teams.

0463. The **main and supporting efforts** operational framework puts emphasis on prioritizing effort among subordinate units. The main and supporting efforts operational framework may be utilized in conjunction with the other two frameworks. Main efforts are subordinate units with preponderance of combat power necessary for mission accomplishment. Supporting efforts are units that support the success of the main effort.

Combined Arms Operations Planning Team

0464. To enable the integration of capabilities, commanders of combined arms team shall incorporate an operational planning team composed of subject matter experts from the different combat arms or competencies that are incorporated in a mission-specific force mix of the combined arms team. The C2 shall integrate capabilities through a comprehensive planning process that ensure efficient, effective and synchronized employment of combat power. This will be further discussed in the later part of the chapter.

Prepare

0465. Preparation consists of activities performed by units to improve their ability to execute an operation. Preparation includes, but is not limited to: plan refinement; rehearsals; intelligence, surveillance, and reconnaissance; coordination; inspections; and, movement. It creates conditions that improve friendly forces' opportunities for success.

0466. Preparation requires staff, unit, and soldier actions. Mission success depend as much on preparation as planning. Rehearsals help staffs, units, and soldiers to better understand their roles in upcoming operations, practice complicated tasks, and ensure equipment and weapons function properly. Activities specific to preparation include:

a. Revising and refining the plan;

b. Conduct rehearsals;

c. Force tailoring and task-organizing;

d. Surveillance and reconnaissance;

e. Training;

f. Troop movements;

g. Pre-combat checks and inspections;

h. Sustainment preparations;

i. Integrating new Soldiers and units;

j. Subordinate confirmation briefs and back briefs.

Rehearsals

0467. A CATm conducts rehearsals to prepare for upcoming operations. A rehearsal is a session in which the commander and staff or unit practices expected actions to improve performance during execution. There are four primary types of rehearsals: back-brief, combined arms rehearsal, support rehearsal, and battle drill or standard operating procedure rehearsal. Methods for conducting rehearsals are limited only by the commander's imagination and available resources. The combined arms commander uses rehearsals as a tool to ensure his staffs and subordinates understand the concept of operations and commander's intent. The extent of rehearsals depends on available time.

0468. A **back-brief** is a briefing by subordinates to the commander to review how subordinates intend to accomplish their mission. This is a result after the commander issues orders to subordinates verbally or through written fragmentary orders. It provides feedback to determine the staff's understanding (FM 6-0) and confirmation brief—briefing subordinate leaders give to the higher commander immediately after the operation order is given. It is their understanding of his intent, their specific tasks, and the relationship between their mission and the other units in the operation (ADRP 5-0). Commanders conduct back-briefs and confirmation briefs between themselves and within staff elements to ensure mutual understanding. Support rehearsals help synchronize each warfighting function with the BCT's overall operation

0469. The **combined arms rehearsal** ensures that subordinate plans synchronize with those of other units, and that subordinate commanders understand the intent of higher headquarters. The execution matrix, decision support template, and operations order outline the rehearsal agenda.

0470. CATms conduct **reconnaissance and security rehearsal** to ensure correct information. The documents needed include the information collection matrix, the reconnaissance and security overlay, and the enemy situation template.

0471. The **fire support rehearsal** focuses on maximizing the ability of fire support systems to support the maneuver plan, and achieve the commander's intent. The fire support rehearsal (including any augmenting fire support from the division artillery or a field artillery brigade) may be used to prepare for a combined arms rehearsal or it may be used after a combined arms rehearsal to refine and reinforce key fire support tasks.

0472. The combined arms **sustainment rehearsal** ensures the synchronization of sustainment efforts before, during, and after combat operations. The sustainment rehearsal validates the "who, what, when, where, and how" of support.

Execute

0473. Executing is putting the plan into action by applying combined arms to accomplish the mission and using situational understanding to assess progress and make execution and adjustment decisions. It focuses on concerted action to seize, retain, and exploit the initiative.

0474. During execution, commanders at all levels locate themselves where they can exercise command and sense the operation. Sometimes this is at the tactical operations center (TOC). At other times, commanders may use the command group or mobile TOC to command from a forward location. Commanders balance the need to make personal observations, provide command presence, and sense the mood of subordinates from a forward location with the ability to maintain communication and control with the entire force from a command post. No matter where they are located, commanders are always looking beyond the current operation to anticipate future actions. They must periodically step back from the current situation and look at how the force is positioning itself for future operations.

0475. In execution, the battle staff-primarily through the Operations Planning Team (OPT) —integrates forces and warfighting functions to accomplish the mission. The staff assesses short-term actions and activities as part of this integration. While the CS (E-XO) integrates staff activities among all operations planning and functional cells, the operations officer integrates the operation through the operations planning cell. Other staff principals integrate within their areas of expertise.

0476. Staffs help commanders control the integrating processes and continuing activities during execution. In addition, commanders, assisted by their staffs, perform the following activities specific to execution:

a Focus assets on the decisive operation.

b Adjust CCIRs and EEFIs based on the situation.

c Adjust control measures.

d Manage movement and positioning of supporting units.

e Adjust unit missions and tasks as necessary.

f Modify the concept of operations as required.

g Position or relocate committed, supporting, and reserve units.

Assess

0477. Assessing is the continuous monitoring and evaluation of the current situation, particularly the enemy, and progress of an operation. Commanders, their staff and subordinate commanders continuously assess the current situation and the progress of the operation and compare it with the concept of operations, mission, and commander's intent. Commanders of combined arms teams monitor the situation for unanticipated successes, failures, or enemy actions. They look for opportunities, threats, and acceptable progress. Throughout the operation, commanders visualize, describe, and direct changes to the operation. Assessments allow the commanders to adjust accordingly to the fluidity of the operational environment.

0478. Assessment precedes and guides every operations process activity and concludes each operation or phase of an operation. It involves a comparison of forecasted outcomes to actual events. Assessment entails three tasks:

a. Continuously assessing the enemy's reactions and vulnerabilities.

b. Continuously monitoring the situation and progress of the operation towards the commander's desired end state.

c. Evaluating the operation against measures of performance and measures of effectiveness.

Integrating Processes

0479. Certain integrating processes occur during all operations process activities. They must be synchronized with each other as well as integrated into the overall operation:

a. Intelligence Preparation of the Operational Environment (IPOE).

b. Targeting.

c. ISR Synchronization.

d. Composite Risk management.

e. Knowledge Management.

0480. The following activities continue during all operations process activities. They are synchronized with one another and integrated into the overall operation:

a. Intelligence, Surveillance, and Reconnaissance.

b. Security Operations.

c. Protection.

d. Liaison and Coordination.

e. Terrain Management.

f. Information Management.

g. Airspace Command and Control

Combined Arms Tactical Operations Center (TOC)

0481. The Combined Arms Tactical Operations Center (TOC) is the hub of where combined arms integration and synchronization happen. The TOC provides the venue where the commander and his staff conducts the operations process (plan-prepare-execute-assess operations). The combined arms commander reconfigures his staff into an operation planning team, functional cells and elements that will execute the operations process.

Commander and Staff Organization

0482. The commander and staff organization provides a chain of command and control between the commander and his staff, each with a designated responsibility and task. The combined arms commander drives the conduct of operations while the combined arms staff supports the commander, assist subordinate units, and inform units and organizations outside the headquarters. The staff supports the commander's understanding, making and implementing decisions, controlling operations, and assessing progress. The staff makes recommendations and prepares plans and orders for the commander.

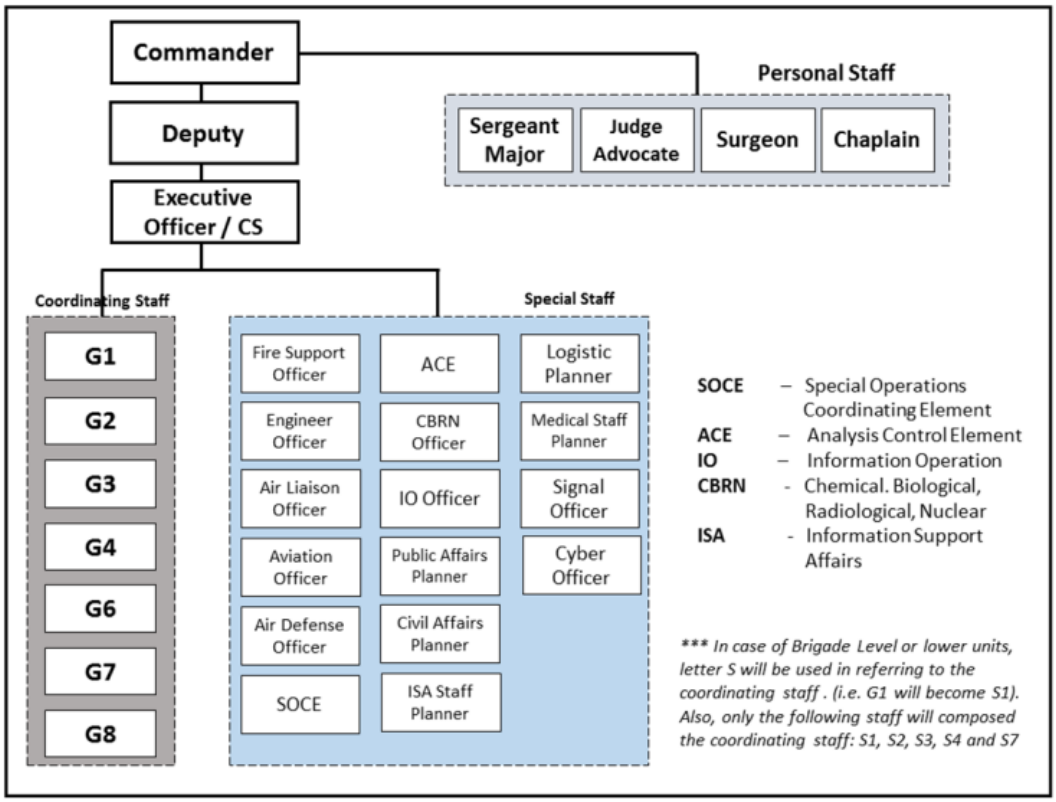


Figure 4.4 - CATm Commander and Staff Organization

0483. **Commander.** Commanders are responsible for all their staffs do or fail to do. A commander cannot delegate this responsibility. The final decision, as well as the final responsibility, remains with the commander. When commanders assign a staff member a task, they delegate the authority necessary to accomplish it. Commanders provide guidance, resources, and support. They foster a climate of mutual trust, cooperation, and teamwork.

0484. **Assistant Division Commander (ADC) or Deputy Brigade Commander (DBC).** The commander determines the duties and responsibilities of the assistant division commanders or deputy brigade commanders. In a division or brigade, the ADC or DBC extends the commander's span of control in areas and functions as the commander designates. The commander may designate assistant commander or deputy as task force commanders depending upon assigned missions. The assistant commander's or deputy specific duties vary from division to division and brigade to brigade.

0485. **Chief of Staff (Executive Officer).** The Chief of Staff (CS) or Executive Officer (Ex-O) is the commander's principal assistant. Commanders normally delegate executive management authority to the CS (Ex-O). As the key staff integrator, the CS (Ex-O) frees the commander from routine details of staff operations and the management of the headquarters or the TOC. Division and higher units are assigned a CS. Brigade and battalions are assigned an Ex-O. The CS (Ex-O) ensures efficient and prompt staff actions.

Coordinating Staff Officers

0486. Coordinating staff officers are the commander's principal staff assistants and are directly accountable to the Chief of Staff (CS) or Executive Officer (Ex-O). Coordinating staff officers are responsible for one or a combination of broad fields of interest. They help the commander develop, coordinate and supervise the execution of plans, operations, and activities. Collectively, through the CS (Ex-O), they are accountable for the commander's entire field of responsibilities.

0487. **Assistant Chief of Staff, AC of S, G1 (S1) Personnel.** The AC of S, G1 (S1) is the principal staff officer for all matters concerning human resources support (military and civilian). The G1 (S1) also serves as the senior adjutant general officer in the command. Specific responsibilities of the G-1 (S1) include manning, personnel services, personnel support, and headquarters management. The G1 (S1) has coordinating staff responsibility for the civilian personnel officer and the equal opportunity advisor. In the operation order or operation plan, the G1 (S1) focuses on the personnel support aspect of Annex F (Sustainment). The G1 (S1) acts as the assistant to the Chief of the Sustainment functional cell of the OPT. Likewise, the G1 (S1) heads the personnel support section under the Sustainment Functional Cell.

0488. **Assistant Chief of Staff AC of S, G2 (S2) Intelligence.** The AC of S, G2 (S2) is the principal staff officer and Chief of the **intelligence warfighting function cell** of the OPT. The G2 (S2) is responsible for providing intelligence to support current and future operations and plans. The G2 (S2) gathers and analyzes information on enemy, terrain, weather, and civil considerations for the commander. The G2 (S2) is responsible for the preparation of Annex B (Intelligence) and assists the assistant chief of staff, operations (G3 [S3]) in the preparation of Annex L (Information Collection). The G2 (S2), together with the G3 (S3), helps the commander coordinate, integrate, and supervise the execution of information collection plans and operations. The G2 (S2) helps the commander focus and integrate these assets and resources to satisfy the battalion through division intelligence requirements.

0489. **Assistant Chief of Staff, AC of S, G3 (S3) Operations.** The AC of S, G3 (S3) operations officer's responsibilities are unique within the coordinating staff. The G3 (S3) is the principal staff officer responsible for all matters concerning training, operations and plans, and the chief of the **movement and maneuver warfighting functional cell.** In addition to coordinating the activities of the movement and maneuver warfighting function, the operations officer is the primary staff officer for integrating and synchronizing the operation as a whole for the commander. While the CS (EX-O) directs the efforts of the entire staff, the operations officer ensures warfighting function integration and synchronization across the planning horizons in current and future operations and plans. Additionally, the operations officer authenticates all plans and orders for the commander to ensure the warfighting functions are synchronized in time, space, and purpose in accordance with the commander's intent and planning guidance. The G3 (S3) has coordinating staff

responsibility for the aviation officer, engineer officer, information operations officer, as well as other staff officers residing in the movement and maneuver cell. In conjunction with other staff, the G-3 (S-3) prepares Annex A (Task Organization), Annex C (Operations), and Annex M (Assessment) to the operation order or operation plan.

0490. The G3 (S3) has responsibilities for plans and operations. Overall, this officer prepares, coordinates, authenticates reviews, publishes, and distributes written operation orders and plans. This includes the command SOP, plans, orders (including fragmentary orders [FRAGOs] and warning orders [WARNOs]), exercises, terrain requirements, and products involving contributions from other staff sections. The G3 (S3) provides coordination, integrates information collection, and allocates resources.

0491. The G3 (S3) may designate a **battle captain** to assist him in supervising the movement and maneuver functional cell. The battle captain may perform as either head of current operations section or the plans section.

0492. **Assistant Chief of Staff, AC of S, G4 (S4) Logistics.** The AC of S, G4 (S4) is the principal staff officer for sustainment plans and operations, supply, maintenance, transportation, and services. At division level, the G4 is titled the chief of sustainment functional cell of the OPT. At brigade level and below, the S4 serves as the principal staff officer coordinating sustainment. The G4 (S4) helps the support unit commander maintain logistics visibility with the commander and the rest of the staff. As the chief of sustainment, the G4 has coordinating staff responsibility for the G1, G8, transportation officer, and the surgeon. The G-4 (S-4) prepares Annex F (Sustainment) to the operation order or operation plan.

0493. **Assistant Chief of Staff, AC of S, G6 (C4S).** The AC of S G6 (S6) is the principal staff officer for all matters concerning command and control, communications, and cyber systems (C4S), network operations, network transport, information services, and spectrum management operations within the unit's area of operations. The G6 (S6) prepares Annex H (Signal) and participates in preparation of Appendix 12 (Cyberspace Electromagnetic Activities) to Annex C (Operations) with input from the G2 (S2) and in coordination with the G3 (S3), to the operation order or operation plan. The G6 (S6) performs as the C4S integrator under the Mission Command functional cell.

0494. **Assistant Chief of Staff, AC of S, G7 (S7) CMO.** The AC of S, G7 (S7) is the principal staff officer responsible for all matters concerning civil military operations. The G7 (S7) is also the Chief of the CMO functional cell of the OPT. The G7 (S7) supervises civil affairs, public affairs and information support affairs, evaluates civil considerations during IPB, mission analysis, and prepares the groundwork for transitioning the area of operations from military to civilian control. The G7 (S7) advises the commander on the military's effect on civilians in the area of operations, relative to the complex relationship of these people with the terrain and institutions over time. The G7 (S7) is responsible for enhancing the relationship between Army forces and the civil authorities and people in the area of operations. The G7 (S7) is required at all echelons from battalion through divisions. The G7 (S7) prepares Annex K (Civil Affairs Operations) to the operation order or operation plan.

0495. **Assistant Chief of Staff, AC of S, G8, Training.** AC of S, G8 (Training Officer) is the principal staff responsible for all matters concerning training. At the brigade and battalion

level, the S3 acts as the Training Officer. The G8 (Training Officer) supervises the training requirements of the command. The G8 (Training Officer) ensures that all training shall focused on the unit's METL.

Special Staff Officers

0496. Every staff has special staff officers. This section addresses the specific duties of each special staff officer. The number of special staff officers and their responsibilities vary with authorizations, the desires of the commander, and the size of the command. If a special staff officer is not assigned, the officer with coordinating staff responsibility for the area of expertise assumes those functional responsibilities. In addition, special staff officers can be pulled from adjacent or other units in order to fill the required expertise for each function. During operations, special staff officers work in parts of the CP designated by the commander, CS, or their supervising coordinating staff officer. In general, the CS (XO) exercises coordinating staff responsibility over those special staff officers without a coordinating staff officer.

0497. **Fire Support Coordinator (FSCOORD)/Fire Support Officer (FSO).** The FSCOORD is the principal staff officer responsible for the fires warfighting function and the **chief of fires functional cell** at division level. At brigade and below, the FSO serves as a special staff officer for fires and chief of the Fires functional cell. This officer synchronizes and coordinates fire support for the S3 who integrates fire support into plans and operations. The chief of fires has coordinating responsibility for the air and missile defense officer and the air liaison officer. The chief of fires or fire support officer prepares Annex D (Fires) to the operation order or operation plan.

0498. **Engineer Officer /Chief of Protection.** The Engineer Officer is the Chief of protection and the principal advisor to the commander on all matters relating to the protection of the force. The chief of protection has coordinating responsibilities all officers/staff related to the protection warfighting function. This includes combat engineering, CBRN, EOD, air defense, K-9, and MP.

0499. The engineer officer resides in the protection cell and is responsible for planning and assessing survivability operations. The engineer officer is also involved in planning and operations under the movement and maneuver functional cell, sustainment functional cell and intelligence functional cell. For example, mobility and counter-mobility are part of movement and maneuver, general engineering is part of sustainment, and geospatial engineering supports the intelligence warfighting function. The engineer officer prepares Annex G (Engineer) to the operation order or operation plan.

04100. **Air and Missile Defense Officer.** The air and missile defense officer is responsible for coordinating air and missile defense activities and plans with the area air and missile defense commander, joint force air component commander, and airspace control authority. The air and missile defense officer coordinates the planning and use of all joint air and missile defense systems, assets, and operations. A key role of the air and missile defense officer is to advise the commander on developing a critical asset list and a defended asset list of assets that can be defended by air and missile defense forces. Army forces air and missile defense plans are synchronized with the area air defense commander's area air defense plan, the joint force air component commander's joint air operations plan and daily air tasking order, and the airspace control authority's airspace control plan, and daily airspace control

order. The air and missile defense officer prepares a portion of Annex D (Fires) to the operation order or operation plan.

04101. **Air Liaison Officer.** The air liaison officer is responsible for coordinating aerospace assets and operations, such as close air support, air interdiction, air reconnaissance, airlift, and joint suppression of enemy air defenses. The air liaison officer is the senior Air Force officer with each tactical air control party.

04102. **Aviation Officer.** The aviation officer is responsible for coordinating Army aviation assets and operations at the different echelons. The aviation officer's responsibilities include, but are not limited to— supervision and training over Army aviation operations; monitoring aviation flying-hour, standardization, and safety programs; planning and supervising Army aviation operations; providing technical advice and assistance on using Army aviation for evacuation (medical or other), and; participating in targeting meetings.

04103. **Special Operations Coordinating Element (SOCE).** The Special Operations Coordinating Element is the focal point for the coordination between SOF elements and other elements of the CATm as well as for the synchronization of special operations activities in the context of Combined Arms Operations. SOCE shall act as the primary adviser to the Commander on all matters pertaining to capabilities, limitations and employment of special operations forces.

04104. **Deployable Intelligence Support Element (DISE).** The mission of the DISE is to perform collection management; produce all-source intelligence; provide IEW technical control; and disseminate intelligence and targeting data. The DISE supports the commander in executing mission command and planning future missions across the range of Army operations. The DISE is the G2's (S2's) primary organization for controlling IEW operations and producing intelligence.

04105. **Chemical, Biological, Radiological and Nuclear (CBRN) Officer.** The chemical, biological, radiological, and nuclear (CBRN) officer is responsible for CBRN operations, obscuration operations, and CBRN asset use. The CBRN officer prepares a portion of Annex C (Operations) and a portion of Annex E (Protection) to the operation order or operation plan.

04106. **Information Operations (IO) Officer.** The information operations officer is the special staff officer responsible for synchronizing and de-conflicting information-related capabilities employed in support of unit operations. The information operations officer is authorized at division through brigade. Coordinated by the G7 (S7), the information operations officer leads the information operations element located in the movement and maneuver cell at echelons above brigade. The responsibilities of the information operations officer include, but are not limited to — analyzing the information environment; identifying opportunities to employ information-related capabilities; assessing the risk associated with the use of information-related capabilities; providing input to the synchronization matrix for the use of available information-related capabilities in support of unit operations; identifying information-related capabilities support gaps not resolvable at the unit level; coordinating with other Army, Service, or joint forces to use information-related capabilities to augment existing unit capability shortfalls; providing information as required in support of operations security at the unit level; providing information as required in support of military deception at the unit level; leading the information operations working group; assessing the effectiveness of employed information-related capabilities.

Personal Staff Officers

04107. The Personal staff officers work under the immediate control of, and have direct access to, the commander. The commander establishes guidelines or gives guidance on when a personal staff officer informs or coordinates with the CS (Ex-O) or other staff members. Some personal staff officers have responsibilities as special staff officers and work with a coordinating staff officer. They do this case-by-case, depending on the commander's guidance or the nature of the task. Personal staff officers also may work under the supervision of the CS (Ex-O).

04108. **Command Sergeant Major (CSM).** The command sergeant major is the senior noncommissioned officer of the command at battalion and higher echelons. CSM out policies and enforce standards for the performance, training, and conduct of enlisted Soldiers. They give advice and initiate recommendations to the commander and staff in matters pertaining to enlisted Soldiers. In operations, a commander employs the command sergeant major throughout the area of operations to extend command influence, assess the morale of the force, and assist during critical events.

04109. **Judge Advocate.** The staff judge advocate is the senior legal advisor in the command and the primary legal advisor to the commander. The commander and the staff judge advocate shall, at all times, communicate directly on matters relating to the administration of military justice, including, but not limited to, all legal matters affecting the morale, good order, and discipline of the command. The staff judge advocate also provides legal advice and support to the staff and coordinates actions with other staff sections to ensure the timely and accurate delivery of legal services throughout the command. The staff judge advocate is a member of the commander’s personal and special staff. A legal support element—typically composed of three judge advocates—deploys in direct support of each brigade-level task force. The staff judge advocate provides complete legal support encompassing the six core legal disciplines—military justice, international and operational law, administrative and civil law, contract and fiscal law, claims, legal assistance. The staff judge advocate prepares a portion of Annex C (Operations) and Annex F (Sustainment) to the operation order or operation plan. It must be noted that the Judge Advocate is only applicable to division sized CATms.

04110. **Surgeon.** The surgeon is responsible for coordinating health assets and operations within the command. This officer provides and oversees medical care to Soldiers, civilians, and enemy prisoners of war. The surgeon prepares a portion of Annex E (Protection) and Annex F (Sustainment) to the operation order or operation plan. Surgeon responsibilities include, but are not limited to— advising the commander on the health of the command; providing health education and training; coordinating medical evacuation, including Army dedicated medical evacuation platforms (air and ground); ensuring the establishment of a viable veterinary services program; ensuring an area medical laboratory capability for the identification and confirmation of the use of suspect biological and chemical warfare agents by opposition forces; establishing clinical laboratory capabilities, including blood banking; planning for and implementing preventive medicine operations; supervising and preparing health-related reports and statistics; advising on the effects of the medical threat on personnel, rations, and water, and; advising how operations affect the public health of personnel and the indigenous populations.

04111. **Chaplain.** The mission of the chaplain is to perform or provide religious support to the unit by assisting the commander in providing for the free exercise of religion and religious, moral, and ethical leadership. Chaplains execute their distinct religious support mission for Soldiers, members of other military services, family members, and authorized civilians in a variety of geographical locations, operational situations, and circumstances. Chaplains provide religious support and advise commanders on the impact of religion, reflecting the dual roles of religious leaders and religious staff advisors. Chaplains and chaplain assistants are assigned at all battalion and higher echelons. The chaplain prepares a portion of Annex F (Sustainment).

Combined Arms TOC Structure and Layout

04112. The combined arms tactical operations center is a unit headquarters where the commander and his staff perform the operations process activities. The combined arms commander organizes the reconfigures the headquarters into a TOC composing of an Operations Planning Team, functional cells and elements with networks, equipment and facilities. There are two (2) types of TOC: the main TOC and the advance TOC. The main TOC is a facility containing the majority of the staff designed to control current operations, conduct detailed analysis, and plan future operations. The advance TOC is a facility containing a tailored portion of a unit headquarters designed to control portions of an operation for a limited time. Shown on figure 4.4 is the typical layout of combined arms Tactical Operations Center (TOC).

04113. The commander and selected staff members comprise a command group. Selected staff members accompany the commander to exercise mission command away from a command post.

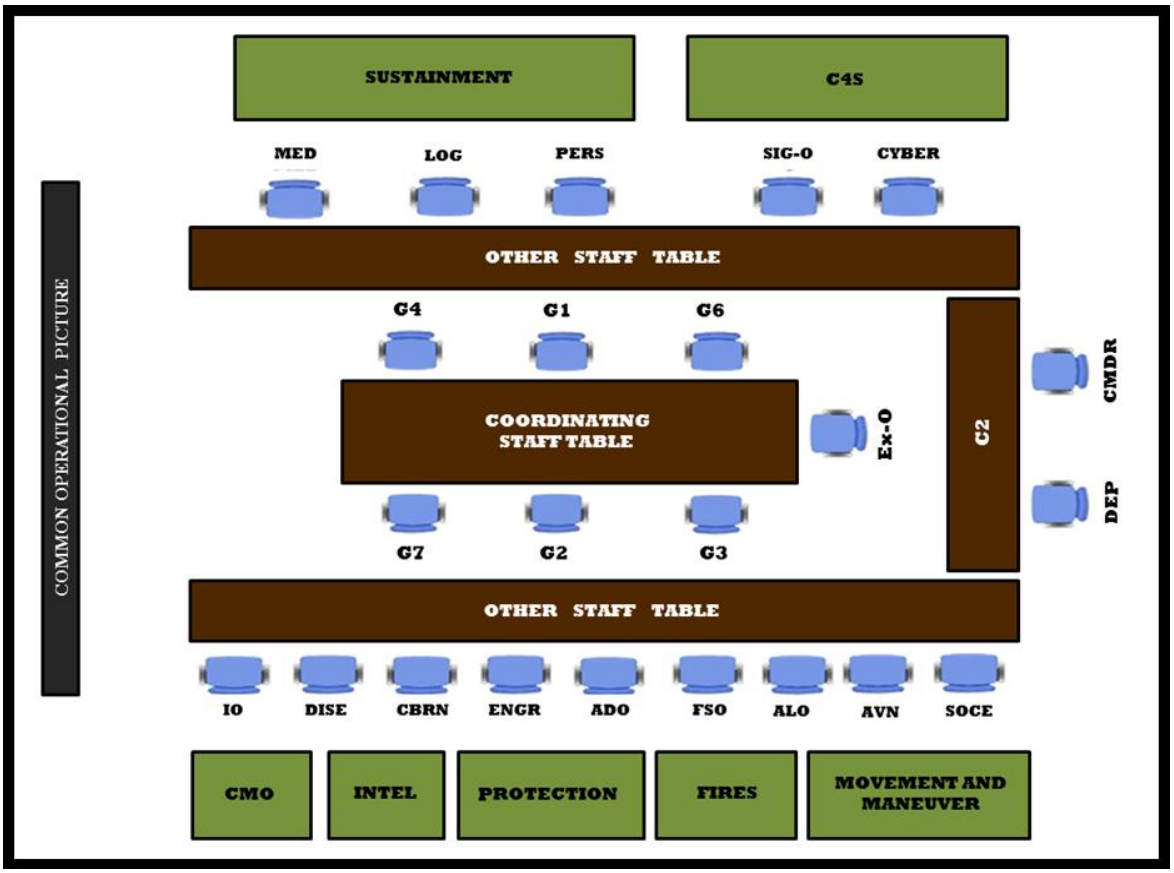


Figure 4.5 - Typical Combined Arms Operation TOC Layout

04114. Depending upon the situation, the combined arms commander reconfigures his staff into an Operations Planning Team (OPT) or as the Battle Staff that would perform operations process in the TOC. Figure 4.5 shows the organizational structure of the OPT in the Tactical Operations Center (TOC).

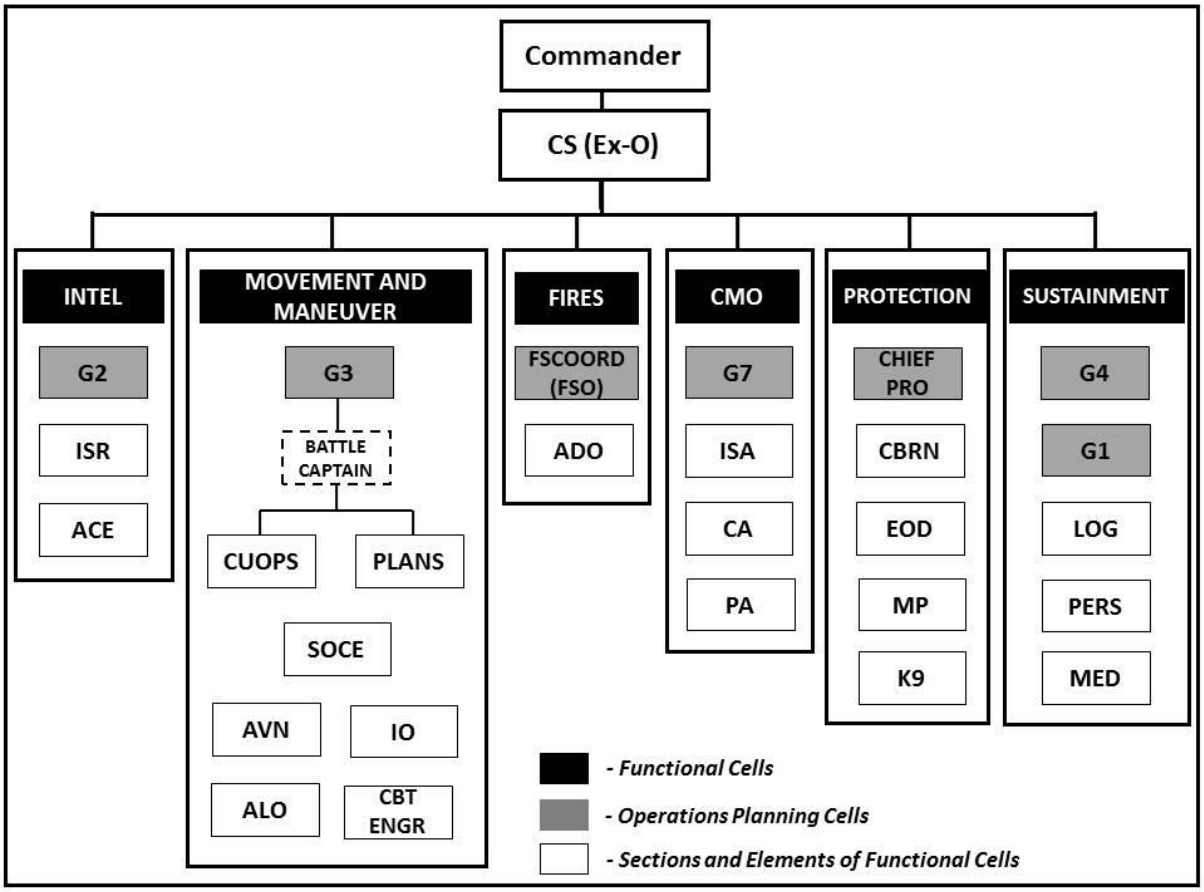


Figure 4.6 – OPT Structure in a TOC

Operations Planning Team

04115. The OPT (battle staff) is divided into types of cells: the Operations Planning Cell and the functional cells (by warfighting functions - intelligence, movement and maneuver, fire support, protection, and sustainment). Operations Planning Cell and functional cells provide staff expertise, communications, and information systems that work in concert to aid the commander in planning and controlling operations. Each cell is composed of staff elements – personnel and equipment – to provide the expertise depending on the field of specialization (Operations Planning or functional).

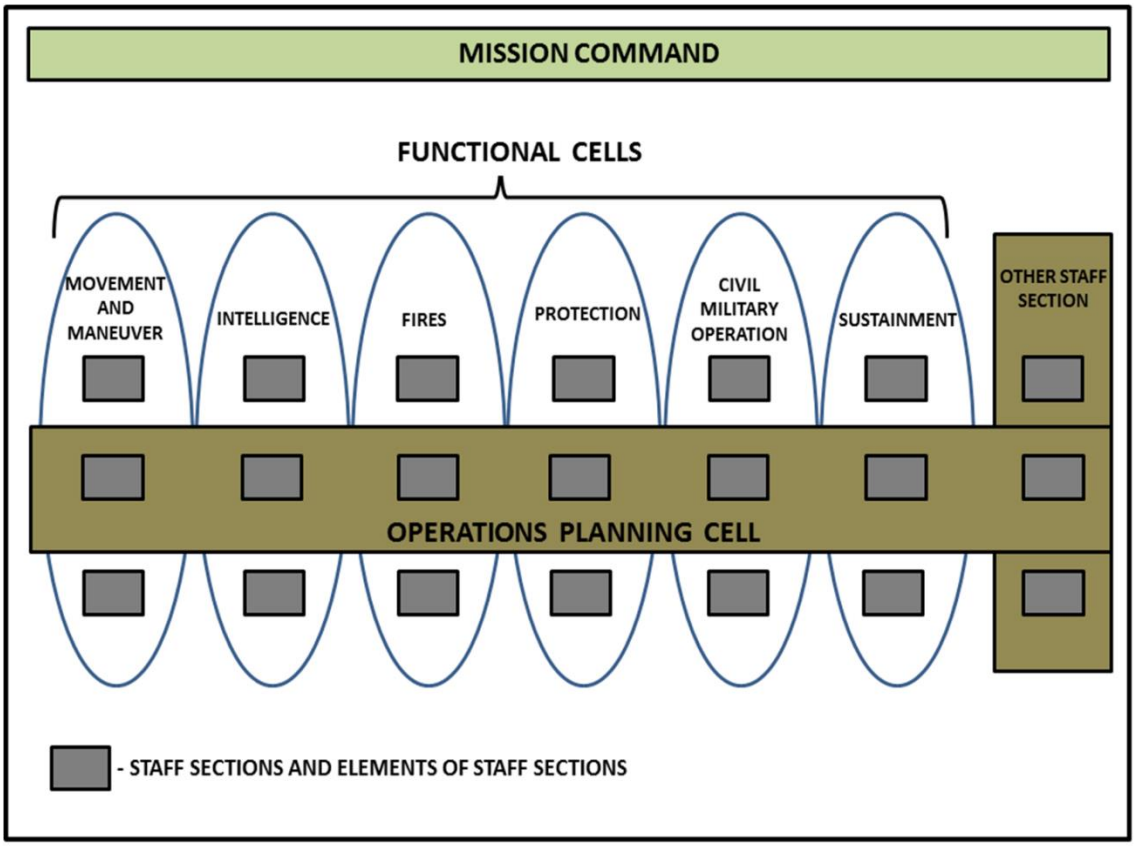


Figure 4.7 – Operations Planning Cell and Functional Cell

Operations Planning Cell (OP Cell)

04116. Cross functional by design, the OP Cell coordinates and synchronizes forces and warfighting functions. It is composed of subject matter experts of the different arms to synchronize and integrate the different army capabilities by incorporating the different staff expertise in the planning process. It facilitates the collaboration among the planning staff, making the planning process more efficient and effective. The OP Cell oversees execution of the current operation. It assesses the current situation while regulating forces and warfighting functions according to the commander's intent and concept of operations. It is also responsible for planning operations. It assists the commanders in the development of plans and orders.

Functional Cells

04117. Functional cells coordinate and synchronize forces and activities by warfighting function. The functional cells within a TOC are movement and maneuver, fire support, intelligence, protection, Civil Military Operations and sustainment.

04118. **Mission Command Cell.** The entire tactical operations center assists the commander in the exercise of mission command. Therefore, commanders do not form a specific mission command functional cell. All of the various TOC cells and staff sections assist the commander with specific tasks of the mission command warfighting function. For example, all functional and operational planning cell assist the commander with conducting the operations process. As such, the tactical operations center as a whole, including the

commander, deputy commanders, and command sergeants major, represents the mission command warfighting function.

04119. **Movement and Maneuver Cell.** The movement and maneuver cell coordinates activities and systems that move forces to achieve a position of advantage. This includes tasks related to gaining a positional advantage by combining forces with direct fire or fire potential (maneuver) and force projection (movement). Elements of the operations, airspace control, aviation, engineer, geospatial information and service, and space support element form this cell. Staff elements in the movement and maneuver cell also form the core of the current operations integration cell. The unit's operations officer leads this cell. When necessary, the Movement and Maneuver Cell may include an SOF representative. The G3/S3 acts as the Chief of Movement and Maneuver functional cell.

04120. **Intelligence Cell.** The intelligence cell coordinates activities and systems that facilitate understanding of the threats, terrain and weather, and other relevant aspects of the operational environment. The intelligence cell requests, receives, and analyzes information from multiple sources to produce and distribute intelligence products. The intelligence cell consists of the majority of the intelligence staff and may include an SOF representative. Higher headquarters may augment this organization with additional capabilities to meet mission requirements. The unit's G-2 (S-2) intelligence officer leads this cell.

04121. **Fires Cell.** The fires cell coordinates, plans, integrates, and synchronizes the employment and assessment of fires in support of current and future operations. The fires cell develops high payoff targets and selects targets for attack. The fires cell recommends targeting guidance to the commander. The fires cell plans, synchronizes, coordinates, and integrates adaptable fires matched to a wide range of targets and target systems. The fires cell coordinates target acquisition, target dissemination, and target engagement functions for the commander. At the division level, the air and missile defense section is integrated within the fires cell to ensure coordination of sense and warning systems, synchronization of fires, and airspace integration. The fires cell coordinates activities and systems that provide collective and coordinated use of Army indirect fires, joint fires, and air and missile defense through the targeting process. The fires cell includes elements of fire support, the air and missile defense section, and liaison officers from joint or multinational fire support agencies. Additional augmentation to the fires cell includes the naval surface fire support liaison officer and Army space support team(s). The unit's chief of fires (or fire support officer at brigade and below) leads this cell.

04122. **Protection Cell.** The protection cell coordinates the activities and systems that preserve the force through risk management. This includes tasks associated with protecting personnel and physical assets. Elements of the following staff sections form this cell: chemical, biological, radiological, and nuclear; engineer; personnel recovery; and provost marshal. Additionally, a safety officer is assigned at theater army and, with augmentation, as required down to the brigade level. The protection cell coordinates with the signal staff section to further facilitate the information protection task. The engineer officer acting as the chief of protection leads this cell.

04123. **Sustainment Cell.** The sustainment cell coordinates activities and systems that provide support and services to ensure freedom of action, extend operational reach, and prolong endurance. It includes those tasks associated with logistics, personnel services, and health service support. The following staff sections form this cell: personnel, sustainment,

financial management, and surgeon. The G4 (S4) acts as the chief of sustainment (or logistics officer at brigade and below) leads this cell.

04124. **Civil Military Operation (CMO) Cell.** The CMO cell coordinates activities and systems that gain popular support, establish legitimacy of action, protect the force, and influence the adversary's cognitive capability. It facilitates the conduct of civil affairs, public affairs and information support affairs. The following staff sections composed the CMO Cell: Civil affairs, public affairs and information support affairs. The G7 (S7) acts as the Chief of the CMO functional cell.

Staff Processes and Procedures

04125. The combined arms team must man, equip, and organize the tactical operations center to control operations for extended periods. The combined arms commander, assisted by his staff, arranges TOC personnel and equipment to facilitate 24-hour operations, internal coordination, information sharing, and rapid decision-making. The commander ensures procedures to execute the operations process within the headquarters enable mission command. The staff uses standard operating procedures, battle rhythm, meetings, and running estimates to assist them in command post operations.

a SOPs

b Battle rhythm

c Meetings

d Running estimate

04126. **SOPs** that assist with effective mission command serve two purposes. Internal SOPs standardize each TOCs internal operations and administration. External SOPs developed for the entire force standardize interactions among TOCs and between subordinate units and TOCs. Effective SOPs require that all soldiers know their duties and train to standards.

04127. A headquarters' **battle rhythm** consists of a series of meetings (to include working groups and boards), briefings, and other activities synchronized by time and purpose. The battle rhythm is a deliberate daily cycle of command, staff, and unit activities intended to synchronize current and future operations.

04128. **Meetings** are gatherings to present and exchange information, solve problems, coordinate action, and make decisions. They may involve the staff; the commander and staff; or the commander, subordinate commanders, staff, and others as necessary (including unified action partners). Who attends depends on the issue. Commanders establish meetings to integrate the staff and enhance planning and decision making within the headquarters.

04129. Commanders also identify staff members to participate in the higher commander's meeting, including working groups and boards. (JP 3-33 discusses the various working groups and boards used by joint force commanders.) Decisions made during meetings must be shared internally within the headquarters, with the higher headquarters, and with subordinate units.

04130. Effective plans and successful executions hinge on accurate and current running estimates. A **running estimate** is the continuous assessment of the current situation used to determine if the current operation is proceeding according to the commander's intent and if planned future operations are supportable. Failure to maintain accurate running estimates may cause errors or omissions resulting in flawed plans or bad decisions during execution. Running estimates are principal knowledge management tools the combined arms commander and staff use throughout the operations process. In their running estimates, the commander and each staff section member continuously considers the effect of new information and update the following: facts; assumptions; friendly force status; enemy activities and capabilities; civil considerations; conclusions and recommendations.

Mission Command System

04131. Commanders cannot exercise mission command alone. The mission command system enhances the commander's ability to conduct operations. Commanders organize a mission command system to—

a Support the commander's decision making.

b Collect, create, and maintain relevant information and prepare knowledge products to support the commander's and leaders' understanding and visualization.

c Prepare and communicate directives.

d Establish the means by which commanders and leaders communicate, collaborate, and facilitate the functioning of teams.

04132. To provide these four overlapping functions, commanders arrange the five components of their mission command system: personnel, networks, information systems, processes and procedures, and facilities and equipment.

Personnel

04133. A commander's mission command system begins with people. Soldiers and leaders exercise disciplined initiative and accomplish assigned missions in accordance with the commander's intent, not technology. Therefore, commanders base their mission command system on human skills, knowledge, and abilities more than on equipment and procedures. Trained Soldiers and leaders form the basis of an effective mission command system; commanders must not underestimate the importance of providing training. Key personnel dedicated to mission command include seconds in command, command sergeants major, and staffs.

Network

04134. Generally, a network is a grouping of things that are interconnected for a purpose. Networks enable commanders to communicate information and control forces whether mounted or dismounted. Networks are key enablers to successful operations. Commanders systematically establish networks to connect people. These connections can be established socially through the introduction of two personnel to perform a task, or technically through

information systems. Commanders develop and leverage various social networks—individuals and organizations interconnected by a common interest—to exchange information and ideas, build teams, and promote unity of effort.

Information Systems

04135. Commanders determine their information requirements and focus their staffs and organizations on using information systems to meet these requirements. An *information system* consists of equipment that collect, process, store, display, and disseminate information. This includes computers—hardware and software—and communications, as well as policies and procedures for their use (ADP 6-0). Staffs use information systems to process, store, and disseminate information according to the commander's information priorities. These capabilities relieve the staff of handling routine data. Information systems—especially when integrated into a coherent, reliable network—enable extensive information sharing, collaborative planning, execution, and assessment that promote shared understanding.

Processes and Procedures

04136. Commanders establish and use systematic processes and procedures to organize the activities within the headquarters and throughout the force. Processes are a series of actions directed to an end state, such as the military decision making process. Procedures are standard, detailed steps, often used by staffs, which describe how to perform specific tasks to achieve the desired end state. For example, the military decision making process provides the commander, staffs, and subordinate commanders an orderly method for planning. Procedures govern actions within the mission command system to make it more effective and efficient. For example, standard operating procedures often provide detailed unit instructions on how to configure common operational picture displays. Adhering to processes and procedures minimizes confusion, misunderstanding, and hesitation as commanders make frequent, rapid decisions to meet operational requirements.

Facilities and Equipment

04137. Commanders systematically arrange facilities and equipment, including command posts, platforms, operation centers, signal nodes, and all mission command support equipment. A facility is a structure or location that provides a work environment and shelter for the other components of the mission command system. Facilities range from a command post composed of vehicles and tentage, to platforms, to hardened buildings. Examples of equipment needed to sustain a mission command system include vehicles, radio or signal equipment, generators, and lighting.

Chapter V
Training For Combined Arms Operations

"The best form of "welfare" for troops is first class training, for this saves unnecessary casualties." - Field Marshal Erwin Rommel

0501. Successful military actions can only be achieved through intensive training. The Army trains its forces to prepare them for their mission and adapt their capabilities to any changes in the operational environment. The readiness of Army forces is paramount at all times and training is the key to maintaining this readiness. As part of a joint force, the Army provides ready forces to combatant commanders that are able and capable to execute various tasks across the spectrum of operations. At the end of the day Army trains to win and training is the most important thing the Army does to prepare for conflict.

0502. The cross training of Special Operations Forces with conventional forces shall also be integrated in Combined Arms Training Exercises in consideration of any geographic security crisis situation of national concern and strategic implication, similar to the Zamboanga siege in 2014 and the Marawi City crisis in 2017.

0503. Training for Combined Arms Operations and all other unit trainings, to be effective, must be tough, relevant, rigorous, realistic, challenging, and properly resourced and battle focused. It must provide the full range of experiences needed to produce agile, adaptive leaders and soldiers, and versatile units.

0504. Complementary to the training of Army forces is the development of Army Leaders (Officers and non-commissioned officer). Competent and confident leaders are essential to unit readiness and successful deployments.

Principles of Combined Arms Training

0505. The Army provides a force commander with combined arms trained and ready force that expands the command's range of military options in operations. A responsive training program will generate a fully integrated and synchronized proper force mix capable of achieving combined arms effects and accomplish the mission. Combined arms training will be anchored on the following principles (see Figure 5.1).

Commanders are Responsible for Training.
NCOs Train Individuals, Crews and Small Teams.
Train Based on METL.
Train Realistically.
Train to Standard Based on Doctrine
Train to Develop Adaptability.
Train to Sustain and Maintain.
Train for Combat Proficiency.
Train in a Multi-Echelon Technique.
Train as a Combined Arms Team

Figure 5.1 - Principles of Combined Arms Training

Commanders are Responsible for Training

0506. Commanders are responsible for ensuring that their units are ready and capable of performing their missions at all times. They are directly responsible, and accountable for all aspect of training. Commanders cannot delegate this responsibility.

Non-Commissioned Officers Train Individuals, Crews, and Small Teams

0507. Non-Commissioned Officers (NCOs) are the primary trainers of enlisted personnel, crews, and small teams. NCOs help officers train units. NCOs develop and conduct training for their subordinates that supports the unit training plan, coach other NCOs, advised senior leaders, and help develop junior officers. The Army must empower its NCOs to be able to develop their competencies being the backbone of the organization.

Train Based on METL

0508. Units trains based on its mission essential tasks (METs). A *mission-essential task* is a collective task on which an organization trains to be proficient in its designed capabilities or assigned mission. Training based on METL is the key to having battle focused training.

Train Realistically

0509. Realism in training will provide the troops firsthand experience on what they will expect in actual combat while on a controlled environment. Realistic training builds competence and confidence among officers and soldiers. To achieve realism, training must be tough and physically and mentally challenging and should all factors that affect the conduct of military operations in the operational environment.

Train to Standard Based on Doctrine

0510. Training must be based on well-defined standards and established doctrines. Leaders know and enforce standards to ensure their organization meets mission requirements. When no standard exists, the commander establishes one and the next higher commander approves it.

Train to Develop Adaptability

0511. Effective leaders understand that change is inevitable in any operational environment. The time to react to change can be short. Hence they trained their units and soldiers to become adaptive. Adaptability comes from training under complex, changing conditions, with minimal information available to make decisions. Critical and creative thinking are key in developing adaptive leaders and soldiers.

Train to Sustain and Maintain

0512. Training prepares units and individuals to be resilient. Training must prepare units and soldiers for the stress of operations. In the same manner, units must conduct maintenance to ensure equipment is serviceable and available for the conduct of training and for mission accomplishment. Soldiers and equipment must always be in a high-state of readiness and must be capable of fighting for sustained periods of time.

Train for Combat Proficiency

0513. The development of combat capable warfighting forces is the goal of combined arms training. To be proficient in the performance of critical tasks and missions, units should practice the performance of this critical tasks and missions. The Army's forces must continuously hone its core competency that is warfighting.

Train in a Multi-Echelon Technique

0514. Multi-echelon training is a training technique that allows for the simultaneous training of more than one echelon on different or complementary tasks. It optimizes training time for subordinates during higher unit training events. Training multiple tasks concurrently preserves valuable time while capitalizing on the opportunity to train related tasks at the same time.

Train as a Combined Arms Team

0515. Combined Arms is the fundamental basis for the organization and operation of Philippine Army forces. It is also the Army's contribution to the Joint Force operations. Combined arms proficiency requires effective integration of all warfighting functions. The standard for effective combined arms training requires continuous execution of both functional and combined arms tasks in order to achieve the desired effects.

Combined Arms Operations Training Cycle

0516. The general objective of Combined Arms Training is to develop combined arms ready Army forces that are capable of accomplishing tasks across the range of Army operations in any operational environment.

0517. Illustrated in Figure 5.2 is the Combined Arms Training cycle which will serve as guide in the planning, execution and evaluation of combined arms trainings.

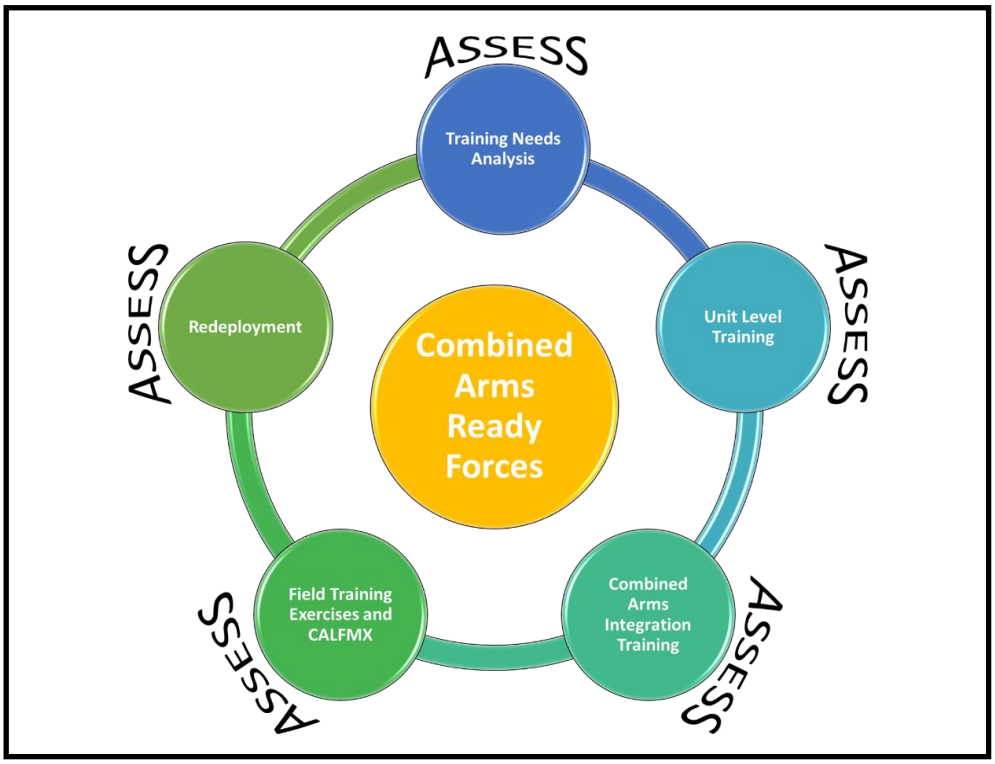


Figure 5.2 - Combined Arms Operations Training Cycle

Training Need Analysis

0518. Training Need Analysis (TNA) is a pre - training process where HPA concerned offices, training planners/training implementers/managers and the unit to be trained identifies and complete all the requirements prior to the conduct of unit sustainment training. This phase begins when the units to train are and ends with the completion of all the requirements based on DOTMLPF.

0519. TNA addresses the competency gaps between what is and what should be, all-encompassing within the contemporary operating environment (COE). It offers solution for certain deficiency or discrepancy that averts the attainment of goals or fulfillment of a desired state of training and force preparedness.

Unit Level Training

0520. Unit Level Training is a process where units train individually based on their respective capabilities/METL in preparation for their integration as part of a combined arms team. Training will include among others, squad/platoon/company sustainment trainings, mission planning (MDMP for the battle staff and TLP for small unit leaders). The training is conducted prior to their movement to the combat readiness training area and is held at the unit's respective home station.

Combined Arms Integration Training (CAIT)

0521. Combined Arms Integration Training is a process by which the elements of the combined arms team are integrated and trained as one unit. CAIT designed to synchronize fires and maneuvers, and ensure the interoperability of the different war fighting functions. Force Integration Training is conducted through levelling briefs, Situational Training/Problem Solving Exercises; Mission Planning Exercises (MDMP/TLP/) and combined arms rehearsals/Rehearsal of Concept (ROC) Drills.

Field Training Exercise (FTX) and Combined Arms Live Fire and Maneuver Exercise (CALFMX)

0522. FTX is military maneuver or simulated wartime operation involving planning, preparation, and execution. It is carried out for the purpose of training and evaluation. It may be a combined, joint, or single-Service exercise, depending on participating organizations. Offensive/Defensive Operations and Support to Civil Authorities are the usual scenarios that the units on training perform. On the other hand, Combined Arms Live Fire and Maneuver Exercise is an event wherein the units on training conduct actual live fire and maneuver in a controlled environment to test and evaluate the effectiveness and interoperability of its weapon system through a scenario – based fire and maneuver. Though CALFMX is important especially for the newly organized CATms, it can be waved as part of the combined arms training depending on the recommendation of the commander of the unit facilitating the exercise. Exercise safety and limitation of resources are among the factors that can cause the cancellation of any live fire exercises.

Redeployment

0523. Redeployment is the process by which units undergone combined arms training are redeployed either in their former area of operations or in new areas where their competencies are required. During this phase, it is expected that trained units are combined arms ready and fully mission capable based on the context of DOTMLPF. In addition, a post training evaluation is conducted one year after the redeployment of the trained units to measure their effectiveness.

Assessment

0524. Assessment is a continuous process that includes evaluating training, conducting an organizational assessment, and preparing a training assessment. It is conducted throughout each process in order to ensure that all identified gaps are addressed before proceeding to the next phase. Assessment is more than just training evaluation, and encompasses a wide variety of inputs. Assessments include such diverse systems as training, force integration, logistics, and personnel. It shall check the responsiveness of training area, facilities, program of instructions, trainers and the capacity of the trained unit to absorb learning. It provides the link between the unit's performance and the Army standard. Evaluation of training is, however, a major component of assessment.

0525. Assessment is a shared responsibility of both the training managers and implementers as well as the Commander of the unit undergoing training in order to ensure the attainment of the training objectives.

Combined Arms Training Design

0526. Combined Arms Training Exercise (CATEX) simulates wartime operations with emphasis on a realistic, battle–focused setting that will hone and train battlefield commanders, staffs and units for Combined Arms Operations. The conduct of CATEX will follow the designed framework illustrated in Figure 5.3.

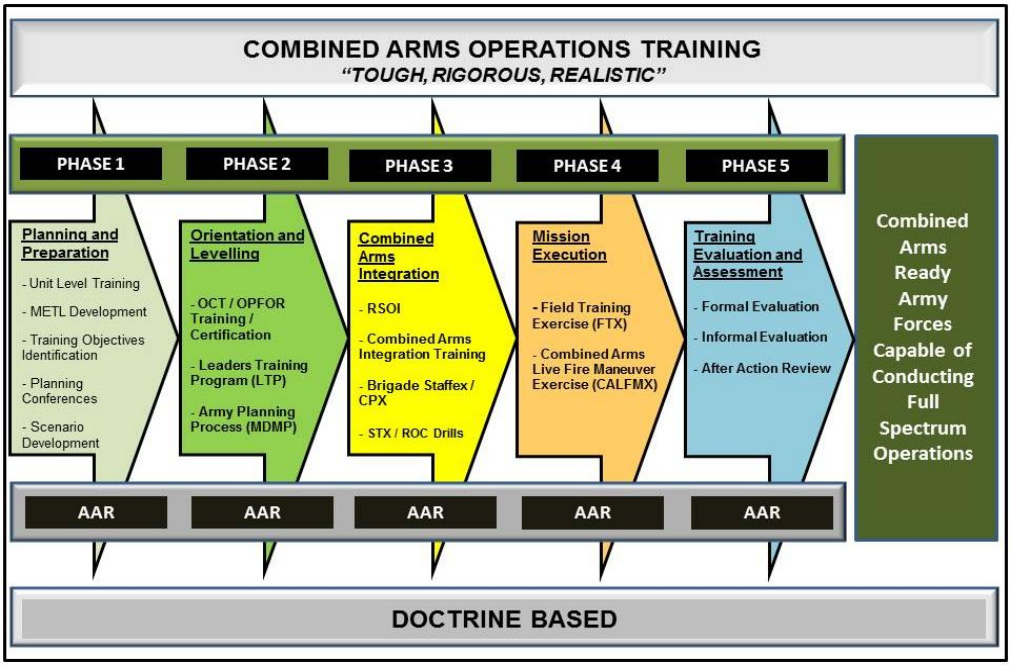


Figure 5.3 - Combined Arms Training Design

0527. CATEX will be conducted in five (5) phases that includes; Phase 1 – Planning and Preparation; Phase 2 – Orientation and Levelling; Phase 3 – Combined Arms Integration; Phase 4 - Mission Execution; and Phase 5 – Training Evaluation and Assessment.

0528. CATEX shall be primarily conducted at a Combat Readiness Training Area were facilities and training environment offers a more realistic battlefield scenario. TRADOC, being the force developer of the Army shall be responsible in administering the conduct of CATEX through the Land Warfare Center.

0529. CATEX may also be conducted at the unit's respective stations in the form of Command Post Exercise (CPX), Staff Exercise (STAFFEX), Tactical Exercise without Troops (TEWT), and other Table Top Exercises (TTX). Unit Commanders are primary responsible in the conduct of these exercises to ensure the readiness of its key staff and units in executing combined arms operations.

CATEX Organizational Structure

0530. Combined Arms Training Exercise shall be managed by different subject-matter experts, intelligently chosen and organized to ensure the achievement of the objectives of the training.

0531. They shall be responsible in the design, planning, execution and overall supervision of the training. They shall conduct evaluation and assessment to provide unit commanders on the performance of the rotational training unit based on the training standards set by the Army. Exercise structure may vary based on the exercise design and number of participants however; constant to any exercise structure is the Exercise Command Group that is composed of Exercise Director and staff, the Exercise Control Group, the Exercise Support Group and the Training Audience/Players composed of the Rotational Training Unit or the Blue Force, and the Opposing Force or the Red Force. Civilian Role Players can be also utilized based on the exercise design. A typical CATEX Organizational Structure is illustrated in Figure 5.4.

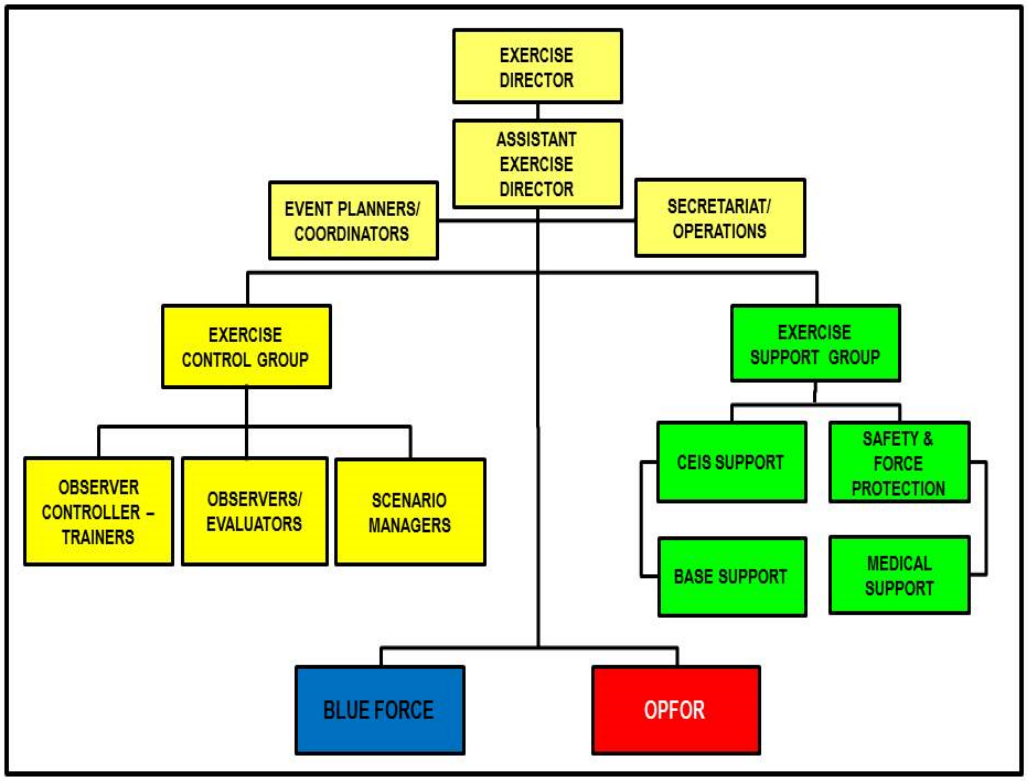


Figure 5.4 - CATEX Organizational Structure

Training Evaluation

0532. Training evaluations are a critical component of any training. Evaluation measures the demonstrated ability of soldiers, commanders, leaders, battle staffs, and units against the Army standard. Evaluation of training is integral to standards-based training and is the cornerstone of leader training and leader development.

0533. All training must be evaluated to measure performance levels against the established Army standard. The evaluation can be as fundamental as an informal, internal evaluation performed by the leader conducting the training. Evaluation is conducted specifically to enable the unit or individual undergoing the training to know whether the training standard has been achieved. Commanders must establish a climate that encourages candid and accurate feedback for the purpose of developing leaders and trained units.

0534. Evaluation of training is not a test; it is not used to find reasons to punish leaders and soldiers. Evaluation tells the unit or the soldier whether or not they achieved the Army standard and, therefore, assists them in determining the overall effectiveness of their training plans. Evaluation produces disciplined soldiers, leaders and units. Training without evaluation is a waste of time and resources.

0535. Leaders use evaluations as an opportunity to coach and mentor subordinates. A key element in developing leaders is immediate, positive feedback that coaches and leads subordinate leaders to achieve the Army standard. This is a tested and proven path to develop competent, confident adaptive leaders.

0536. Evaluations can be informal, formal, internal, external, or any combination, thereof;

a. **Informal evaluations** take place when a leader conducts training with his unit, for example when a squad leader trains his squad to assault an objective. Another example would be whenever a leader visits ongoing training, for instance when a battalion commander observes company training. This type of evaluation provides real time feedback on the training environment and the proficiency resulting from training.

b. **Formal evaluations** are resourced with dedicated evaluators and are generally scheduled in the long-range or short-range training plans. Formal evaluations are normally highlighted during short-range training briefings. To the maximum extent possible, headquarters two echelons higher conduct formal external evaluations; for example, division commanders evaluate battalions, brigade commanders evaluate companies, and battalion commanders evaluate platoons.

c. **Internal evaluations** are planned, resourced, and conducted by the organization undergoing the evaluation.

d. **External evaluations** are planned, resourced, and conducted by a headquarters at an echelon higher in the chain of command than the organization undergoing the evaluation or a headquarters outside the chain of command.

Evaluation System and Quality Control

0537. Evaluation and quality control of the training shall be provided through the following mechanisms:

a. Utilization of trained OCTs;

b. After Action Reviews (AARs) during FTX and CPX;

c. Stressful and realistic exercises;

d. Doctrine-based evaluation checklists during exercises;

e. Conduct of evaluation surveys (during and immediately after the training);

f. Conduct of validation (six months after the occupation of the deployment area);

g. Supervision by the Training Committee during the training and during the validation period; and

h. Commander's responsibility to determine if his unit is trained or not based on the evaluation standards set in accordance with the battalion's METL.

Glossary

Army Power - It is the summation of the potency that any total army forces generates. It is composed of the warfighting functions (mission command, movement and maneuver, fires, intelligence, CMO, protection, and sustainment) enhanced by information and held together by leadership.

Battle Rhythm - It consists of a series of meetings (to include working groups and boards) briefings, and other activities synchronized by time and purpose. It is a deliberate daily cycle of command, staff, and unit activities intended to synchronize current and future operations.

Combined Arms Operations - It is the synchronization, integration and seamless execution of Army capabilities to achieve desired effects.

Combined Arms Operations Process - It provides a broad process in conducting operations, and consists of major command and control activities performed during operations namely: planning, preparing, executing and continuously assessing the operations.

Combined Arms Tactical Operations Center - It is the unit headquarters where the commander and staff perform the operations process activities.

Combined Arms Team – The full integration and application of two or more elements of an Army into an operation. It is composed of combat arms, combat support and combat support that are further categorized into maneuver, maneuver support and combat sustainment components.

Combined Arms Training Exercise - It is a training exercise that simulate wartime operations with emphasis on a realistic, battle-focused setting that will hone and train battlefield commanders, staff, and units for combined arms operations.

Combined Operations - these are operations conducted by two or more nations, usually undertaken within the structure of a coalition or alliance.

Defensive Operations - these are combat operations to defeat an enemy attack, gain time, control key terrain, and economize forces.

Force Tailoring - It is the process of determining the right force mix of units and the sequence of their deployment in support of a joint force commander or a CATm commander.

Full Spectrum Operations - conduct of offensive, defensive, security and support to civil authority operations across the range of military operations simultaneously and normally as an independent force or as part of an interdependent joint force to seize, retain, and exploit the initiative, accept prudent risk, in order to create opportunities and achieve decisive results.

Functional Cell - It is composed of staff sections and elements that coordinate and synchronized forces and activities organized by warfighting functions.

Interagency Operations - these are missions and tasks performed to uphold national sovereignty and interest and support government.

Joint Operations - employment of two or more services in a single operation to achieve operational and strategic objectives.

Landpower Maneuver - It is the Philippine Army's land defense strategy that provides an overarching construct on how the Army, as part of the Joint Force, will conduct prompt and sustained land operations in order to secure the nation's sovereignty and land territory.

Main Effort - it is the unit, battle positions, sectors, zone, axis, avenue of approach, are of operations, theater operations, and so forth, the senior commander has determined as the most important task and purpose at that time, whose success will make the most difference in the accomplishment of the higher commander's overall mission or objective.

Meetings - These are gatherings to present and exchange information, solve problems, coordinate action, and make decisions. They may involve the staff, commander and staff, or the commander, subordinate commanders, staff, and others as necessary.

Mission-essential task - This is a collective task on which an organization trains to be proficient in its design capabilities or assigned mission.

Mission Command – It is the army's preferred method of command and control to ensure success of combined arms operations across the Full Spectrum Operations. It is the exercise of centralized planning and decentralized execution through mission orders.

Offensive Operations - these are combat operations to defeat and destroy enemy forces and seize terrain, resources, and population centers.

Operational Art - It is the cognitive approach by commanders and staff supported by their skill, knowledge, experience, creativity, and judgement to develop strategies, campaigns, and operations to organize and employ military forces by integrating ends, ways, and means.

Operations Planning Cell - Cross functional by design, it coordinates and synchronizes forces and warfighting functions. It is composed of subject matter experts of the different arms to synchronize and integrate the different army capabilities by incorporating the different staff expertise in the planning process.

Operations Planning Team - It is composed of cells, staff sections and elements providing staff expertise, communications, and information systems that work in concert to aid the commander in planning and controlling operations.

Reserve - this is a portion of the force withheld from action or uncommitted to a specific course of action, so as to be available for commitment at the decisive moment.

Running Estimate - It is a continuous assessment of the current situation used to determine if the current operation is proceeding according to the commander's intent and if planned future operations are supportable.

Security Operations - These are combat operations to protect populations, forces and infrastructures, and activities.

Support to Civil Authorities - Support given to civil authorities encompasses military missions, tasks, and activities in support to civil authorities to maintain or re-establish a safe and secure environment, provide essential government services, emergency infrastructure reconstruction, and humanitarian relief.

Supporting Effort - It directly or indirectly creates a necessary condition for the main effort to succeed.

Task Organizing - It is the art of designing an operating force, support staff, or logistics package of specific size and composition to meet a unique task or mission.

Warfighting Functions - It is a group of tasks and related systems (people, organizations, information, and processes) united by a common purpose that commanders use to accomplish missions and training objectives.

Acronym and Abbreviation

AAR	After Action Review
ADA	Air Defense Artillery
ADCON	Administrative Control
ALO	Air Liaison Officer
BCT	Brigade Combat Team
C4S	Command and Control, Communications, and Cyber Systems
CAIT	Combined Arms Integration Training
CALFMX	Combined Arms Live Fire and Maneuver Exercise
CATEX	Combined Arms Training Exercise
CATm	Combined Arms Teams
CBRN	Chemical, Biological, Radiological, and Nuclear
CCIR	Commanders Critical Information Requirements
CMO	Civil Military Operations
CNN	CPP-NPA-NDF
COA	Course of Action
CS	Chief of Staff
CPX	Command Post Exercise
CSM	Command Sergeant Major
DISE	Deployable Intelligence Support Element
DS	Direct Support
EEFI	Essential Elements of Friendly Information
EOD	Explosive Ordnance Disposal
FSCOORD	Fire Support Coordinator
FSO	Fire Support Officer
FTX	Field Training Exercise
GESCON	Graphical Estimate of the Situation in Counterinsurgency
GS	General Support
GSR	General Support Reinforcing
HADR	Humanitarian Assistance and Disaster Response
IO	Information Operations
IPOE	Intelligence Preparation of the Operational Environment
ISR	Intelligence, Surveillance, and Reconnaissance
JIIM	Joint, Interagency, Intergovernmental, Multinational
MDMP	Military Decision Making Process
MET	Mission-Essential task
METL	Mission-Essential Task List
METTC	Mission, Enemy, Terrain, Troops, Civil considerations
MP	Military Police
NCOs	Non-commissioned officers
OCT	Observer-Controller Trainer
OP Cell	Operations Planning Cell
OPCON	Operational Control
OPSEC	Operational Security
OPT	Operations Planning Team

PEFTOK	Philippine Expeditionary Force to Korea
R	Reinforcing
SOCE	Special Operations Coordinating Element
SOF	Special Operations Forces
SOP	Standard Operating Procedures
SOTF	Special Operations Task Force
TACON	Tactical Control
TLP	Troop Leading Procedures
TNA	Training Needs Analysis
TOC	Tactical Operations Center
TOE	Table of Organization and Equipment

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